Still Echoing: Welfare Reform and the Noncognitive Skill Formation of the Next Generation

Kingsford Onyina†

Click here for the latest version

Abstract

How did welfare reform affect the children who grew up under it? The 1990s overhaul of U.S. cash assistance replaced Aid to Families with Dependent Children (AFDC) with Temporary Assistance for Needy Families (TANF), introducing work requirements, time limits, sanctions, and greater state discretion. Prior research has examined effects on adults and children's short-run outcomes, but less is known about whether early-life exposure shaped human capital as exposed cohorts entered young adulthood. I study noncognitive skills (NCS), an overlooked socioemotional dimension of human capital, using validated Big Five-based measures of personality. Linking the PSID core survey to the Child Development Supplement and Transition into Adulthood Supplement, I measure exposure from conception through age five and estimate a triple-difference design exploiting variation across states, birth cohorts, and mothers' likelihood of being affected. I find that early-life exposure lowered young-adult NCS by 0.23 standard deviations among individuals born to likely affected mothers. The decline is concentrated in agreeableness and openness, largest for earliest exposure, and most detectable among girls and White children, though subgroup differences are imprecise. Mechanism results point less to household income or maternal employment and more to early caregiving conditions. The findings show that safety-net reforms can leave lasting marks on less visible dimensions of children's human capital.

†PhD Candidate in Public Policy Analysis at the University of North Carolina at Charlotte. Email: konyina@charlotte.edu. **JEL Codes:** I38, J13, J24, H53.

"Skills beget skills."
— James J. Heckman

1 Introduction

In the 1990s, the United States fundamentally restructured cash assistance for low-income families with children. Through state welfare waivers and the 1996 Personal Responsibility and Work Opportunity Reconciliation Act (PRWORA), Aid to Families with Dependent Children (AFDC), a federal entitlement, was replaced by Temporary Assistance for Needy Families (TANF), a block grant program built around work requirements, time limits, sanctions, and expanded state discretion. The reform was designed to reduce welfare dependence, increase employment, and encourage family formation among the primarily single-mother households served by AFDC (Ziliak, 2015). In doing so, welfare reform changed more than the rules governing benefit receipt. It changed the economic and caregiving environments in which low-income children were raised. By altering maternal employment, income stability, parental time, exposure to sanctions, and household stress, the reform changed several inputs into child development in potentially offsetting ways. Increased work could raise earnings, strengthen routines, and expose children to stronger labor-market attachment. But time limits, sanctions, unstable low-wage work, and the loss of an entitlement floor could also increase hardship, reduce parental time, and heighten stress in already constrained households. The expected effect on children was therefore ambiguous ex-ante.

The central unresolved question is not simply whether welfare reform changed families in the 1990s, but whether those changes left durable developmental marks on children exposed during the earliest years of life. Early studies focused mainly on the first generation, the mothers and families directly subject to the new rules, with attention to employment, welfare participation, income, marriage, fertility, and material hardship (Blank, 2002; Grogger, 2003; Ziliak, 2015). A related literature examined how welfare reform affected children and adolescents, studying outcomes such as cognitive development, schooling, health, behavior, family environment, delinquency, risky behaviors, and adolescent transitions (Bastian et al., 2021a,b; Herbst, 2017; Kalil et al., 2023). This evidence shows that welfare reform mattered for the next generation. Yet the timing of the existing evidence leaves an important question open. Studies of young children generally measure outcomes while children are still young, whereas studies that follow cohorts into early adulthood often examine individuals who were adolescents when reform occurred. Nearly three decades after PRWORA, the empirical window is different. The cohorts who spent their earliest years under the reformed welfare regime have now reached young adulthood, making it possible to ask whether early-childhood exposure produced developmental effects that persisted beyond childhood and adolescence.

Recent work has begun to take this long-run, second-generation perspective, studying how childhood exposure to welfare reform affected educational attainment, food insecurity, family formation, and related outcomes (Corman et al., 2022; Hartley et al., 2022; Vaughn, 2023). This paper builds on that shift by examining a margin of human capital that remains underexplored in this literature: noncognitive skills (NCS). NCS encompasses socioemotional traits and behavioral dispositions, such as agreeableness, conscientiousness, emotional stability, and openness, that shape how individuals regulate behavior, relate to others, and respond to constraints. Scores of literature have documented how NCS independently are powerful determinants of economic mobility, educational attainment, and health (Esping-Andersen and Cimentada, 2018; Heckman and Mosso, 2014; Humphries and Kosse, 2017).

Crucially, NCS are not fixed traits but malleable skills and responsive to environmental condition and shapeable through quality parenting, stimulation, and caregiving well into adolescence (Kautz et al., 2014). This malleability is precisely what makes welfare reform a consequential context for NCS formation: the reform restructured the parenting conditions, household stability, and caregiving environments through which these traits are forged during the most sensitive developmental window. We cannot infer the effects on NCS from findings on cognitive skills, as these distinct dimensions have different malleability periods and respond differently to environmental influences. As Heckman (2000) argues, preoccupation with cognitive test scores at the exclusion of social adaptability and motivation introduces a serious bias into the evaluation of human capital interventions, and this study fills precisely that gap by assessing how the reform shaped this critical component of human capital in the long run. The timing of this analysis is uniquely suited to the question: these children are now young adults whose socioemotional traits have crystallized into stable, measurable patterns (Bleidorn et al., 2022).

To answer this question, I use the Panel Study of Income Dynamics (PSID), linking the Child Development Supplement (CDS) to the Transition into Adulthood Supplement (TAS). These data follow children observed during welfare reform period into young adulthood and allow me to combine information on childhood family structure, maternal education, state of residence, and adult socioemotional outcomes. I measure exposure to welfare reform from conception through age five, the period in which changes in household resources, parental time, and caregiving routines are likely to be especially consequential for later skill formation (Cunha and Heckman, 2007). The empirical strategy exploits variation in the timing and intensity of AFDC waivers and TANF implementation across states, differences in exposure across birth cohorts, and differences in the likelihood of being affected by the reform based on family background. Specifically, I estimate a continuous-dose triple-difference design comparing children born to low-educated single mothers, the group most likely to be directly affected by welfare reform, to children from less affected family backgrounds, across states and birth cohorts.

I find that early-life exposure to welfare reform lowered noncognitive skills in young adulthood by about 0.2 standard deviations among individuals born to mothers most likely to be affected by the reform. The decline is concentrated in agreeableness and openness, with little evidence of changes in conscientiousness, extraversion, or neuroticism. The age pattern is consistent with a developmental interpretation, with the largest effects arising from exposure in the earliest years of life and smaller effects at older ages. Mechanism results suggest that welfare reform operated less through household income or maternal employment than through changes in early caregiving conditions. The effects are most visible among girls and White children, although the differences across groups are not precisely estimated. I then examine whether these socioemotional changes are reflected in adult economic outcomes. In descriptive Mincer-style regressions, agreeableness and openness are negatively associated with earnings, while conscientiousness and emotional stability are positively associated. Since the reform reduced traits that are not rewarded in the labor market, a simple decomposition does not predict an earnings loss and may even imply a gain. Consistent with this, I find no detectable effects on the same cohort's earnings or employment. The results suggest that welfare reform left persistent marks on socioemotional development, but along margins that standard economic outcomes may fail to capture.

These findings contribute to three strands of research. First, the paper extends the literature on the second-generation effects of welfare reform. Existing studies show that childhood exposure to welfare reform affected later outcomes such as educational attainment, fertility, family formation, and other measures of adult socioeconomic status (Bastian et al., 2021a; Dave et al., 2012; Hartley et al., 2022; Herbst, 2017; Vaughn, 2022). I add to this literature by studying noncognitive skills in young adulthood, a dimension of human capital that has received limited attention in evaluations of welfare reform. The results show that the reform's long-run effects on the next generation were not confined to conventional outcomes such as schooling, employment, or family structure, but also extended to socioemotional traits that shape how individuals regulate behavior, interact with others, and navigate adult roles.

Second, the paper contributes to the broader literature on early-life environments and human capital formation. A central implication of skill-formation models is that early investments and family conditions can have lasting effects because skills are self-productive and because early capabilities affect the productivity of later investments (Cunha and Heckman, 2007; Heckman et al., 2013). Consistent with this framework, I show that policy-induced changes in children's early household environments can be reflected in noncognitive skills measured many years later. The results therefore provide evidence that early-life exposure to public policy can leave persistent marks on socioemotional development, complementing evidence on the long-run effects of early childhood conditions on health, education, and adult socioeconomic outcomes (Bennhoff et al., 2024; Currie and Almond, 2011; García et al., 2023; Walker et al., 2022).

Third, the paper informs the literature on safety-net design and child development. Prior work on safety-net programs and long-run noncognitive outcomes has focused largely on unconditional cash transfers or in-kind supports (Akee et al., 2018; Baker et al., 2019; Bolbocean and Tylavsky, 2021; Milligan and Stabile, 2011). Welfare reform provides a distinct policy environment because it combined income support with work requirements, time limits, sanctions, and expanded state discretion. The findings show that the design of safety-net programs, not only the level of transfers, can shape the developmental trajectories of children in low-income families. This distinction is important for policy evaluation because reforms that alter both household resources and behavioral incentives may generate effects that are not fully captured by short-run measures of income, employment, or program participation.

The remainder of the paper proceeds as follows. Section 2 reviews the empirical evidence on welfare reform and children's human capital development. Section 3 presents the conceptual framework. Section 4 describes the data, exposure measure, and empirical strategy. Section 5 reports the main results. Section 6 discusses the findings and concludes.

2 Empirical Evidence

Whilst the impact of the reform on the immediate beneficiaries has been well established, documenting large and well-identified impacts on maternal employment, welfare caseloads, and household income (Blank, 2002, 2009; Grogger and Karoly, 2005; Ziliak, 2015). The impact on the second generation has received limited attention, particularly the human capital development of the children of the affected mothers. Here, I stratify the existing evidence on it's human capital impact into cognitive and noncognitive.

The foundational insight of the cognitive strand comes from experimental evidence. Morris et al. (2009), synthesizing seven welfare experiments conducted across the United States in the 1990s, find that programs simultaneously boosting maternal employment and raising household income through earnings supplements improved young children's achievement scores by approximately 7 percent of a SD, with effects concentrated among children aged five and under. Programs mandating work without supplementing earnings produced no corresponding cognitive gains, establishing that material stability rather than maternal employment per se is the operative developmental input. Quasi-experimental work using variation in state-level reform timing confirmed and extended these signals. Welfare reform improved test score performance among low-income students, with gains concentrated in states experiencing the greatest caseload reductions (Miller and Zhang, 2009), and structural simulation evidence reinforces the mechanism, combining time limits with work requirements increased cognitive attainment by altering maternal employment and welfare use decisions (Chyi et al., 2014; Chyi and Ozturk, 2013). These cognitive improvements carried forward into longer-run human capital accumulation, manifesting in higher enrollment rates, lower dropout rates, and greater college completion among exposed children (Dave et al., 2012; Offner, 2005; Vaughn, 2022). Where adverse effects emerge they trace the same mechanism in reverse, welfare receipt is associated with cognitive score declines operating through maternal stress and income pathways under unstable employment (Heflin and Acevedo, 2011; Herbst, 2017), and attainment effects vary by gender and policy bundle (Bastian et al., 2021a).

Across this strand, the evidence is coherent and conditional, reform moved cognitive and educational trajectories when and because it altered household material conditions. What it cannot tell us is whether the reform shaped the formation of socioemotional traits that are distinct from cognitive ability, differently determined, and independently consequential for long-run human capital.

Complementary strand of the literature motivated by the same query on how the welfare reform altered the household conditions of early childhood, those alterations should have registered not only in test scores but in how the socioemotional and behavioral development of the affected children directly and indirectly. In similar fashion to the cognitive literature, experimental evidence on programs incorporating welfare reform features produced heterogeneous effects on externalizing behaviors, with improvements in some contexts and adverse effects in others, depending critically on whether earnings supplements accompanied the work mandate (Chase-Lansdale et al., 2011; Duncan et al., 2007; Gennetian and Miller, 2002; Gennetian et al., 2004). Chase-Lansdale et al. (2011) find that youth whose mothers increased employment net of welfare participation were less likely to exhibit serious behavior problems, reinforcing the interpretation that income stability, not employment itself, drives behavioral outcomes. The quasi-experimental literature sharpened this mechanism by organizing behavioral outcomes around the nature of the maternal transition rather than the fact of employment. Where transitions off welfare were abrupt or precarious, children exhibited elevated aggressive and withdrawn behavior (Hofferth, 1999). In contrast, where transitions were gradual and combined welfare receipt with work, both internalizing and externalizing problems declined (Dunifon et al., 2003), and stable maternal employment during early childhood predicted further reductions in externalizing behaviors without any corresponding cognitive gains (Pilkauskas et al., 2018). Aughinbaugh and Gittleman (2004) reinforce the conditionality of these effects by finding no impact of maternal employment on risky behaviors, confirming that work alone, absent the stability and income gains that make employment consequential, leaves behavioral trajectories unmoved. In a parallel study, Herbst (2017) exploiting the variation in AYCE lever of the welfare reform observed no impact on affected children's socioemotional development proxied by their externalizing behavior, prosocial behavior and approaches to learning.

Among adolescents, studies indirectly measuring the reforms impact on NCS via engagement in risky and antisocial behaviors also present nuanced findings. Exploiting variation in the implementation of the welfare reform across states and over time, Corman et al. (2017) finds that adolescent aged 15-17 experienced a reduction in minor crimes by 9-11 percent, however, using similar identification strategy in a companion paper they find no effect of the reform on drug related arrests, but they do find some modest increase in arrest. Shifting from high-stake outcomes Dave et al. (2021) also found that the welfare reform rather led to an increase in engagement of risky behavior among adolescents, with the effect being more pronounced for boys as against girls. Recent work by Reichman et al. (2023), extending the outcome set to positive health and social behaviors, find no robust effects on any prosocial dimension. This set of literature also shows how the welfare reform affected the behavioral and socioemotional component of the children and adolescents of the welfare families, thereby directly and indirectly show the noncognitive impact of the reform. However, these findings are largely contemporaneous, capturing the immediate reactions to the reform rather than the enduring NCS that crystallize across early childhood and persist into adulthood.

Across both strands, the literature converges on a finding that is as consistent as it is incomplete. The impact of the welfare reform had substantial impact the human capital of the affected children and adolescents, but the effect was heterogeneous and in every case conditioned by the quality and stability of the household environment the reform created. When reform improved material conditions, cognitive scores rose and behavioral problems fell. When it destabilized them, the reverse obtained. This pattern is not a contradiction, it is the same mechanism operating across different outcome domains, pointing consistently toward the early-life household environment as the operative channel through which welfare reform reached the next generation.

What that convergent finding cannot resolve is whether welfare reform altered the generative layer through which all of these observable outcomes are produced. Both strands measured what children achieved or exhibited rather than the socioemotional traits that produced those outcomes. Critically, much of the existing evidence is contemporaneous, capturing children's outcomes in the years surrounding the reform rather than the socioemotional traits that become more stable and consequential as exposed cohorts enter early adulthood. To my knowledge no study has exploited a validated psychometric measures of NCS to assess how the reform conditioned socioemotional development into young adulthood. This gap is consequential as evidence indicates that children raised in poverty and disadvantaged environments carry lasting NCS deficits into adulthood (Heckman, 2008; Phillips and Shonkoff, 2000). A reform that moved those inputs at scale during the critical early-life window did not merely shift test scores and behavioral responses; it may have recalibrated the socioemotional foundations from which those outcomes derive. That possibility has not been examined, and this paper provides the first direct evidence on it.

3 Theoretical Model

Understanding whether welfare reform shaped the noncognitive development of exposed children requires a framework that links policy-driven changes in early-life environments to the long-run accumulation of socioemotional skills. The relevant question is not whether reform altered parental behavior in the abstract, but whether it moved the specific inputs through which NCS formation proceeds during the developmental window when those inputs are most consequential. This section formalizes that link, derives the theoretical predictions the empirical design tests, and identifies the channels through which welfare reform plausibly altered the early-life environment.

3.1 Skill Formation Technology

I adopt the multistage skill formation model of Cunha and Heckman (2007). Let $t \in \{0, 1, 2\}$ denote prenatal and infancy ($t = 0$), early childhood ($t = 1$), and young adulthood ($t = 2$). Let σ_{it} represent individual i's NCS stock at stage t. Skills evolve through a recursive production technology:

$$\sigma_{i,t+1} = f^t(\sigma_{it}, I_{it}, B_i, \nu_{it}) \tag{1}$$

where B_i captures stable family background factors, I_{it} represents parental investments and home environments during stage t, and ν_{it} is an idiosyncratic shock. Two properties of this technology structure the empirical interpretation.

Self-productivity implies that early skills are persistent:

$$\frac{\partial f^t}{\partial \sigma_{it}} > 0 \tag{2}$$

so early socioemotional traits reinforce themselves across developmental stages. A child who enters early childhood with stronger self-regulation and emotional stability is better positioned to benefit from subsequent parental and institutional investments.

Dynamic complementarity implies that early skill stocks raise the productivity of later investments:

$$\frac{\partial^2 f^t}{\partial \sigma_{it}\, \partial I_{it}} > 0 \tag{3}$$

The stage-specific sensitivity embedded in this technology implies that the in-utero to age-five window is uniquely consequential. Neuroscientific and developmental evidence establishes that the prefrontal cortex, the neurological substrate of self-regulation and emotional control, undergoes its most rapid development during this period, making socioemotional traits particularly responsive to environmental inputs before age five and substantially less malleable thereafter (Cunha and Heckman, 2007; Kautz et al., 2014; Phillips and Shonkoff, 2000). Addressing deficits that emerge in this window at later stages is a poor remediation strategy: the complementarity structure implies that early shortfalls in σ_{i1} reduce the productivity of all subsequent investments, so the costs of environmental deprivation during early childhood compound rather than dissipate over the life course.

Iterating equation (1) yields a cumulative expression for young-adult NCS:

$$\sigma_{i2} = h(B_i, \sigma_{i0}, \bar{I}_{i0}, \bar{I}_{i1}) \tag{4}$$

which establishes that persistent differences in young-adult NCS reflect the joint history of early inputs and family background. Any policy that shifts I_{it} during the early childhood window therefore has the potential to alter the socioemotional trajectory from which all subsequent human capital accumulation derives.

With some assumptions, equation (4) can be taken to the data. I assume that the production function g is linear in inputs and constant across time and individuals. Young-adult NCS, σ_{i2}, are proxied by validated Big Five personality measures from the PSID TAS, which capture the stable socioemotional traits that crystallize by young adulthood

(Bleidorn et al., 2022). Maternal education serves as a proxy for the initial skill endowment σ_{i0}, reflecting the family background conditions that shape the early developmental environment. Early childhood exposure to welfare reform, constructed as the share of the in-utero to age-five window spent under post-reform rules, proxies for family investment decisions I_{it}, capturing the policy-induced variation in early-life inputs that the triple-difference design exploits.

Welfare reform enters this framework by altering I_{it}, the early investments and home environments that govern NCS formation. Let $W_{s,t}$ denote the welfare regime in state s and year t. Children of low-educated single mothers, the group most directly exposed to AFDC waiver and TANF provisions (Bitler and Hoynes, 2010), experienced substantial shifts in both material and psychosocial conditions as states moved from unconditional cash assistance toward a work-conditioned, time-limited regime. These shifts operated through two intertwined channels, each with distinct sub-mechanisms.

The economic investment channel operates through the household's disposable income, material stability, and maternal time. Welfare reform consistently increased labor supply among single mothers (Grogger and Karoly, 2005; Ziliak, 2015), and where employment brought earnings gains, it improved material conditions and reduced income volatility (Schoeni and Blank, 2000). Because family resources and economic stability shape children's emotional regulation, conscientiousness, and behavioral self-control (Akee et al., 2018; Fletcher and Wolfe, 2016; Morris et al., 2009), these gains predict improved socioemotional development. Yet the same rise in maternal employment reduced the parental time available for caregiving, so the net prediction is ambiguous, turning on whether income gains offset the cost of reduced parental presence. Household composition provides a related channel. By changing the costs and resources associated with additional children, welfare reform could alter the number of young children in affected households. A young sibling may dilute per-child material resources and parental attention, while also increasing maternal time demands, stress, and the difficulty of maintaining stable care arrangements. During early childhood, these pressures can reduce the consistency and emotional quality of care received by the index child, thereby weakening the environment that supports NCS formation.

The caregiving-disruption channel operates through parental stress, emotional availability, and the continuity of care, the inputs most directly implicated in NCS formation during early childhood. Time limits, sanctions, and work requirements changed the conditions under which affected mothers provided care, and by drawing maternal time into work, the arrangements through which children received it. To the extent these provisions raised economic insecurity or displaced maternal time from the home, they could reduce the consistency and emotional quality of caregiving, weakening the environment that supports socioemotional development (Heflin and Acevedo, 2011; Kalil et al., 2023). Because early socioemotional development depends on the continuity and relational quality of care, a shift toward non-parental arrangements, even formal ones, may disrupt it, a consideration that distinguishes socioemotional from cognitive development, for which such care is often beneficial. This channel is also distinct from the economic-investment channel, since it can operate even when income is unchanged: material stability does not guarantee a stable or emotionally available caregiving environment. For the in-utero to age-five window in particular, this is the more proximate mechanism, as infants and toddlers are sensitive to the consistency and quality of care rather than to household income directly. The net effect is theoretically ambiguous, since added structure and routine could also improve the home environment for some families.

These two channels can be summarized by a policy-to-investment mapping:

$$I_{it} = g^t(W_{s(i),t}, B_i, u_{it}) \tag{5}$$

where u_{it} captures unobserved household heterogeneity and $s(i)$ denotes child i's state of residence. Because $W_{s,t}$ shifted discretely across states and years as states adopted AFDC waivers and implemented TANF at different points in time, children born in different states and cohorts experienced different shares of their prenatal-to-age-five window under the reformed regime. This discrete cross-state and cross-cohort variation in $W_{s,t}$ generates the exposure index that serves as the identifying source of variation in the empirical design.

The framework generates three predictions. First, under self-productivity and dynamic complementarity, children exposed during the in-utero to age-five window should exhibit measurably different young-adult NCS than otherwise similar children with less early exposure, insofar as reform altered the inputs governing NCS formation during that window. Second, the sign is theoretically indeterminate: if the economic-investment channel dominates, NCS should be higher; if the caregiving-disruption channel dominates, lower. The net effect is an empirical question, depending on which channel prevails. Third, under the critical-period logic, effects should be largest for exposure earliest in the window and attenuate with the age at exposure.

Anchored in this framework, the triple-difference design identifies the plausibly causal effect of welfare-reform-induced variation in early inputs I_{it} on young-adult NCS σ_{i2}. The resulting estimates capture how the institutional restructuring of cash assistance during the most sensitive developmental window shaped the long-run socioemotional development of the cohort it touched.

4 Data

This study leverages the PSID dataset, the world's longest-running nationally representative household panel, along with its CDS and TAS. The PSID began in 1968 with roughly 18,000 individuals in about 4,802 households and has since followed original sample members and their descendants annually through 1997 and biennially thereafter, collecting detailed socioeconomic, demographic, and geographic information. The CDS and TAS were designed to facilitate research on the intergenerational well-being of children and young adults within their family, school, and neighborhood contexts. The CDS, launched in 1997, sampled children born between 1984 and 1997 who were ages 0–12 at baseline. Up to two children per core family were randomly selected and interviewed, alongside their primary caregivers, and were followed until they reached age 18. Upon turning 18, these individuals transitioned into the TAS. The first TAS wave began in 2005 and continues to track these same respondents biennially through approximately age 28, gathering information on educational trajectories, labor-market experiences, social attitudes, and psychological and socio-emotional well-being. Together, the PSID and CDS–TAS structure provides a rare longitudinal framework linking childhood environments to early-adult outcomes within the same individuals. The structure of the PSID-CDS-TAS sample is shown in Figure A1.

The PSID is particularly well suited for this study for four reasons. First, the CDS baseline coincides with the 1990s welfare reform era coupled the kids birth year and month, enabling credible identification of children developmentally exposed to the policy. Second, the PSID provides geocoded state identifiers, allowing linkage to AFDC waiver and TANF implementation timing and facilitating cross-state comparisons. Third, its intergenerational design yields rich information on both children and parents, supporting detailed controls for household socioeconomic conditions and parental behaviors. Fourth, TAS includes widely used and psychometrically validated measures of NCS, the primary outcomes of interest (Goldberg, 1993; Kautz et al., 2014). The PSID and its supplements have been extensively used to examine the intergenerational consequences of early-life policy exposure and environmental shocks (e.g., Dore et al., 2025; Corman et al., 2022 ; Hoynes et al., 2016; Vaughn, 2023). As with most long-running panels, the PSID experiences differential nonresponse and attrition, and oversamples low-income families; however, these features are mitigated using the survey weights provided. Finally, for the welfare reform timing, I extract the data from US Department of Health and Human Services and (Crouse, 1999). Childhood level state macroeconomic variables were also extracted from the University of Kentucky Center for Poverty Research Welfare Data UKCPR, 2026

4.1 Measures

4.2 Welfare Reform Exposure

Following the exposure-share approach of Hoynes et al. (2016), I construct a continuous measure of early-life exposure to welfare reform, defined as the fraction of months between in-utero and age five during which a child resided in a state operating either an AFDC waiver or TANF. I consider the in-utero period because prior scholarship has documented how welfare reform affected maternal prenatal decisions, investments, employment, and stress, shaping the early developmental environment (Kaestner and Chan Lee, 2005). I first assign each child's policy environment based on their birth state and the month-year in which that state first implemented an AFDC waiver or TANF. I convert these dates into monthly format and compute exposure relative to the child's birth month-year. For each cohort, I measure exposure over a 69-month window (9 months in utero plus 60 months after birth). Let B_s denote the child's birth state, t_b the birth month-year, and t_s the first reform month-year for state s. The exposure window spans $[t_b - 9,\ t_b + 60]$. This measure takes the value of 0 for children born entirely before any reform or in states that adopted only after the child aged out of the window, and 1 for children fully exposed from conception to age five. For children whose exposure falls between fully treated and untreated, I quantify dosage as the proportion of the first 69 months of life, from in-utero to age five, spent in a state operating under a major waiver or TANF,

$$\text{Exposure}_i = \frac{\text{Months exposed in state } B_s}{69}.$$

Because all reforms occurred prior to data collection, this exposure is treated as time-invariant. This construction improves upon simple contemporaneous reform indicators by capturing the intensity and timing of early-life policy exposure rather than relying solely on residence during a single survey year. Consistent with prior work (Bitler and Hoynes, 2010), welfare reform is expected to primarily affect children of single mothers with a high school education or less. I therefore define this group as the policy-relevant (target) population. The comparison group includes children of single mothers with more than a high school education and those raised in two-parent households.

4.3 Noncognitive Skills

I measure NCS using validated items from the PSID-TAS data spanning 2013 to 2023, which capture the five Big Five personality domains, namely Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism (Goldberg, 1993; Kautz et al., 2014). Openness reflects curiosity and receptiveness to new ideas; Conscientiousness captures diligence and reliability; Extraversion reflects sociability and external engagement; Agreeableness gauges prosocial orientation; and Neuroticism captures emotional instability and susceptibility to stress. The survey items corresponding to each domain are listed in Table A4. To construct a composite NCS index, I follow the approach of Kling et al. (2007). First, all items are recoded so that higher values consistently reflect more desirable traits. Next, I compute item-level z-scores by subtracting the comparison-group mean and dividing by the comparison-group standard deviation, expressing each trait in standard-deviation units relative to the untreated distribution. Finally, I take the mean of all available standardized items for each respondent. This aggregation improves reliability, reduces measurement error, and avoids the arbitrary weighting and multiple-testing concerns inherent in domain-specific analyses. I then replicate the same procedure for each subscale.

4.4 Covariates

The analysis includes a rich set of controls to account for demographic composition, family background, and state-level policy environments relevant to children's early-life conditions. At the individual level, I adjust for sex, age in adulthood, a nonwhite indicator (with White as the reference), and an indicator for being raised by a grandparent to capture alternative caregiving arrangements. Maternal education and maternal age are included as core measures of socioeconomic status and parental human capital. To account for pre-reform economic and policy conditions in the child's state, and to mitigate bias arising from nonrandom post-reform migration, I incorporate a set of pre-reform state characteristics, namely the unemployment rate, log TANF and log SNAP benefit levels, the poverty rate, the minimum wage, and the generosity of the Earned Income Tax Credit (EITC), each interacted with birth year to allow their influence to vary across cohorts. These factors jointly capture variation in state economic opportunities and safety-net generosity during the birth-cohort window, helping isolate the effect of early-life welfare reform exposure from broader policy and labor-market environments.

4.5 Identification Strategy

To estimate the long-run effect of early-childhood exposure to welfare reform on NCS formation, I adopt a cohort-based triple-difference design that adapts the early-life exposure-share approach of Hoynes et al. (2016). The design exploits three sources of variation: the staggered timing and intensity of AFDC waivers and TANF implementation across states, differences in exposure across birth cohorts, and differences in the likelihood of being affected by the reform based on family background. The interaction of these three dimensions is what identifies the causal effect.

The reduced-form baseline specification linking young-adult NCS to early-life reform exposure is

$$NCS_{isct} = \delta_1 RE_{isc}^{\text{IU-5}} + \delta_2 T_i + \delta_3 \left(RE_{isc}^{\text{IU-5}} \times T_i \right) + \beta X_{isct} + \alpha_s + \gamma_c + \eta_t + \varepsilon_{isct} \tag{5}$$

where i denotes the individual, s the birth state, c the birth cohort, and t the survey year. The outcome NCS is the standardized NCS index. RE is the continuous exposure measure, the share of the in-utero to age-five window spent under a state AFDC waiver or TANF, so the design is a continuous-dose rather than a binary treatment. T equals one for children of low-educated single mothers, the population most likely to be affected by the reform, and zero otherwise. X is a vector of child, maternal, and pre-reform state controls; α_s, γ_c, and η_t are birth-state, birth-cohort, and survey-year fixed effects. Standard errors are clustered at the state level, since welfare reform was assigned by state and its timing and provisions are common to all children within a state, inducing within-state correlation in the errors. Given the limited number of clusters(36), I additionally report wild cluster bootstrap p-values.[1]

The coefficient of interest is δ_3, the triple interaction. It identifies the effect of a marginal increase in the share of the early-life window spent under reform, for likely-affected children relative to the comparison group. The constituent terms have distinct interpretations. δ_1 captures the association between exposure and outcomes for the comparison group ($T = 0$), children not from low-educated single-mother households. Because prior work establishes that reform eligibility was concentrated among low-educated single mothers while marriage and higher maternal education largely limited it (Bitler and Hoynes, 2010; Dave et al., 2021), a coefficient near zero on δ_1 is itself a placebo test: it indicates that exposure carries no signal for children the reform should not have reached. In a robustness specification, I assess robustness across models that sequentially add sets of the control variables and that include state-specific linear cohort trends ($\alpha_s \cdot c$), which allow all states, including early- and late-reform states, to follow differential systematic trends across birth cohorts. However, I am careful in interpreting estimates from models that include state-specific linear cohort trends, because those controls may capture part of the treatment effect in addition to unobserved factors when the treatment response is dynamic (Wolfers, 2006).

Welfare reform was a permanent, non-reversible policy shift, so a child exposed during early childhood was necessarily exposed in later childhood as well. There is therefore no clean "ever-exposed" versus "never-exposed" contrast to exploit. Rather than comparing treated and untreated individuals, the design identifies the effect of additional exposure during the earliest developmental window, from in-utero to age five, conditional on

[1]Following Roodman et al. (2019), the wild cluster bootstrap resamples cluster-level residuals to deliver p-values that remain valid when clusters are few, where conventional cluster-robust standard errors tend to over-reject. I use 9,999 replications with Rademacher weights, clustering on state.

subsequent exposure later in childhood. This is the appropriate estimand given the policy's structure, and it aligns with the developmental hypothesis that the earliest years are the most sensitive period for noncognitive skill formation.

4.6 Identifying Assumption

The key assumption for the DDD estimate to recover an unbiased causal effect is that absent welfare reform, unobserved time-varying state factors would have affected the target and comparison groups similarly. The target group is children of unmarried mothers with at most a high school education, the population for whom cash-assistance eligibility and work requirements were most binding. The comparison group is children of more-educated single mothers and children in two-parent households, who were at much lower risk of welfare receipt and thus far less likely to be affected by reform. Because both groups experienced the same state and cohort environments, differencing against the comparison group nets out any shock common to all children in a state, and identification requires only that no unobserved factor specific to low-educated single-mother families moved with reform timing.

The validity of this approach depends on welfare reform being unrelated to unobserved state-level factors associated with the outcomes, and on similar pre-reform trends for the target and comparison groups. I assess the role of unobserved time-varying state factors and reform-timing endogeneity by adding pre-reform state characteristics, the unemployment rate, poverty rate, TANF and SNAP caseloads, minimum wage, and EITC generosity, interacted with birth cohort, which addresses the possibility that reform timing was itself dependent on a state's economic conditions and welfare history. I further include state-specific linear cohort trends, which allow all states, whether early or late reformers, to follow differential systematic trends across the sample. Finally, I expand the baseline DDD into an event-study framework that decomposes the policy effect into leads and lags, allowing me to formally test for differential trends between the target and comparison groups prior to reform.

4.7 Threats to Identification

4.8 Differential pre-trends

The identifying assumption fails if the target and comparison groups were already on diverging trajectories before reform. To probe this, I expand the baseline DDD into an event-study specification, replacing the single exposure dose with a set of indicators for years relative to state reform adoption, each interacted with target status, with the year immediately preceding adoption omitted as the reference:

$$NCS_{isct} = \sum_{k \neq -1} \pi_k \left(D_{sc}^{k} \times T_i \right) + \beta X_{isct} + \alpha_s + \gamma_c + \eta_t + \varepsilon_{isct},$$

where D_{sc}^{k} indicates that cohort c in state s falls k years from adoption. The pre-adoption coefficients $\{\pi_k\}_{k<-1}$ trace the target-comparison gap in the cohorts before reform could bind, allowing me to test directly for differential trends between the groups and to date the onset of any effect relative to exposure.

4.8.1 Endogenous reform timing

If states timed reform adoption in response to economic or social conditions that also shaped children's development, adoption timing would be endogenous to the outcome. I address this in two ways. First, I regress reform timing, measured as months from January 1992 to the earlier of a state's waiver or TANF adoption, on a vector of pre-reform state characteristics: log population, the unemployment rate, log TANF and SNAP caseloads, the poverty rate, the minimum wage, and the state EITC rate. I estimate this across the full universe of reform states and in specifications that add Census-region fixed effects, apply population weights, and exclude influential states, examining whether adoption timing is systematically related to state conditions. Second, I include these pre-reform characteristics, interacted with birth cohort, directly in the estimating equation, so that any residual correlation between reform timing and state conditions is absorbed rather than left to the error term.

4.8.2 Selective migration

Because exposure is assigned by state of birth, the estimates could be biased if families relocated across states in response to reform, severing the link between assigned and realized policy exposure. To address this, I re-estimate the baseline specification on the subsample of stayers, children whose birth state equals their state of residence, for whom exposure is measured without migration error, building the specification up sequentially from fixed effects through the full control set within this sample. Comparing the exposure-target interaction across the full and stayer samples isolates whether selective migration contributes to the estimated effect.

4.8.3 Negative weighting under staggered adoption

Because states adopted reform at different times and the treatment effect may vary across cohorts and states, the two-way fixed-effects estimator can assign negative weights to some comparisons, allowing already-treated cohorts to act as controls for later-treated ones (de Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021). Since every state in the sample adopts reform between 1992 and 1996, no never-treated group is available as a clean control. To gauge the severity of this concern, I decompose the two-way fixed-effects estimand into its underlying group-by-period treatment effects and examine the associated weights. To guard against the problem directly, I then estimate a stacked DDD in which each adoption cohort c forms a separate sub-experiment, treated units are children in states adopting in year c, and controls are drawn from not-yet-treated states, those adopting strictly after c. Each sub-experiment is trimmed to a common event window and stacked, and the baseline triple difference is estimated on the pooled sample with fixed effects interacted with the sub-experiment index e,

$$NCS_{isce} = \delta_1 RE_{isc}^{\text{IU-5}} + \delta_2 T_i + \delta_3 \left(RE_{isc}^{\text{IU-5}} \times T_i\right) + \beta X_{isce} + \alpha_{se} + \gamma_{ce} + \eta_{te} + \theta_{Te} + \varepsilon_{isce}.$$

The estimator is identical to the baseline; only the sample changes. Because the sub-experiment-specific fixed effects α_{se}, γ_{ce}, η_{te}, and θ_{Te} confine each comparison to within a sub-experiment, treated cohorts are contrasted only against not-yet-treated states, so the design does not draw on the already-treated comparisons that give rise to negative weights.

5 Results

5.1 Descriptives

Table 1 reports survey-weighted means and SD for the analysis variables, and Table A2 compares the target and comparison groups. The average child spent roughly 21 of the 69 months in the in-utero to age-five window under reformed welfare rules, though exposure varies widely across cohorts and states, with a standard deviation of about 22 months spanning children with almost no exposure to those exposed across the entire window. About 15 percent of the sample falls in the target group, children of low-educated single mothers. As expected by construction, the groups differ most on the dimensions that define the target. Target children are far more likely to have a mother without a high school diploma, at 45 percent against 10 percent among the comparison group, and to be nonwhite, at 63 percent against 20 percent, both differences large in standardized terms. The pre-reform state characteristics are comparable across groups, with small, standardized differences for most and only the poverty rate showing a nontrivial gap, at 14.6 against 13.3. That target and comparison families faced broadly similar pre-reform state environments supports the identifying assumption that reform exposure is not standing in for divergent state economic conditions.

[Table 1 About Here]

Appendix Figure A4 plots kernel densities of NCS for the two groups across the composite index and the five subscales. The composite distributions are nearly coincident, while the subscales separate to differing degrees, with agreeableness and openness showing the clearest divergence. These are raw, unconditional comparisons that reflect group differences in background as much as any effect of reform, and the triple-difference design isolates the latter, to which I turn below.

Figures A2 and A3 document the rollout of reform. No state operated an AFDC waiver before 1992, and adoption rose sharply thereafter, so that by PRWORA's enactment in 1996 most states had an active reform, with all states covered by 1997. This staggered adoption generates the cross-state, cross-cohort variation in early-life exposure that identifies the estimator. Figure A2 shows that early adopters were geographically dispersed across the South, Midwest, and West rather than concentrated in a single region, mitigating the concern that the variation reflects regional economic trends rather than welfare policy.

5.2 Baseline Results

I present the main estimates in Tables 2 through 5 using the triple-difference design described in Section 4. Two features of the specification are important for interpretation. First, IU-Age 5 measures the share of the in-utero to age-five window that a child spent under welfare reform. Its interaction with target status therefore captures the effect of moving from no exposure to full exposure during early childhood for children born to mothers most likely to be affected by the reform. Because all NCS outcomes are standardized, the coefficients are interpreted in standard deviation units. Second, treatment is assigned at the state-by-birth-cohort level rather than by realized welfare participation. The estimates are therefore best interpreted as intent-to-treat effects of early-life exposure to the reformed welfare regime among children born to mothers most likely to be affected. They capture the effect of policy exposure, not the effect of actual welfare receipt.

[Table 2 About Here]

Table 2 develops the composite estimate across five specifications of increasing richness. Column 1 absorbs only the fixed effects; columns 2 through 4 add maternal characteristics, young adult controls, and pre-reform state economic conditions interacted with birth cohort; and column 5 adds state-specific linear birth-cohort trends. Consistent with my treatment of the trend specification in Section 4, I take column 4 as preferred, since the linear cohort trends in column 5 risk absorbing part of a dynamic treatment response. The estimate is remarkably insensitive to this sequence. IU-Age 5 x Target enters at −0.23 SD with fixed effects alone and holds at −0.23 SD through the full control set, moving only trivially once trends are added. That the coefficient is unmoved by the progressive inclusion of maternal, child, and state covariates is the first indication that it reflects the reform rather than the correlation of exposure with background. The coefficient on IU-Age 5 alone, which recovers exposure among the comparison group, is a precise zero throughout, as expected for children the policy was not designed to reach.Because state clusters are few, conventional standard errors could overstate precision, so I report wild cluster bootstrap p-values throughout; the DDD estimates retain significance under this stricter standard.

[Table 3 About Here]

Table 3 then decomposes the composite into its five subscales under the preferred specification, and the decline proves selective rather than uniform. It concentrates in agreeableness and openness, at −0.31 SD and −0.44 SD, both significant at five percent. The remaining traits, conscientiousness (−0.16 SD), extraversion (−0.07 SD), and neuroticism (−0.14 SD), are smaller and fall short of significance, though all five estimates share the same negative sign. Because the subscales form a family of five related hypotheses, I report Benjamini et al. (2006) sharpened two-stage q-values controlling the false discovery rate at 0.10. Agreeableness and openness each survive the correction at a q-value of 0.04; the remaining three do not and the remaining three do not, and these two subscales likewise retain significance under the wild cluster bootstrap.

[Table 4 About Here]

Table 4 estimates the specification separately by child sex. The effect is carried by girls, whose total effect is −0.26 SD on the composite at one percent and −0.37 SD on agreeableness at five percent, while the estimates for boys are muted and statistically indistinguishable from zero. This asymmetry warrants care. The triple interaction, IU-Age 5 x Target x Female, is itself insignificant, so although the effect is precisely estimated among girls, the data do not support the stronger claim that it exceeds the effect for boys. I read the estimates as marking where the effect is identified rather than as a contrast between the sexes, and I hold to that distinction throughout the heterogeneity analysis.

[Table 5 About Here]

Table 5 repeats the exercise by child race, and the pattern parallels the one for sex. The declines concentrate among White children, at −0.29 SD on the composite, −0.47 SD on agreeableness, and −0.61 SD on openness, each significant at five percent or better, while the total effects for nonwhite children are smaller and never significant. As before, the triple interaction is insignificant, so I take these estimates to locate the subgroup in which the effect is most precisely estimated rather than to establish a genuine difference between White and nonwhite children.

5.3 Mechanisms

I examine the channels through which reform exposure operates using the 1997 Child Development Supplement. One caveat frames what follows. The supplement records household and caregiving conditions at a single point rather than continuously across the exposure window, so these variables capture a snapshot of the early environment around 1997 rather than the full trajectory from in-utero to age five. They are best read as indicators of which inputs the reform moved, not as a complete time-path of exposure. Table 6 reports triple-difference estimates for each candidate channel, spanning childcare arrangements, household income, maternal employment, maternal distress, the presence of a young child, and mother-child closeness.

[Table 6 About Here]

The estimates point more strongly to the caregiving environment than to household economic resources. Reform-exposed target children are 8 percentage points more likely to be in formal childcare and 11.4 percentage points more likely to live with another young child in the household, while informal care is unchanged. The economic margins tell a different story. Household income barely moves and is imprecisely estimated, and maternal employment is effectively zero. These estimates provide little evidence that the reform raised the material resources that might have buffered children. Maternal distress is positive but imprecise, and mother-child closeness is unchanged.

A feature of the policy lets me probe the caregiving reading more directly. If the harm works by displacing early maternal care, it should fade where the reform left that care intact, and states differed sharply in exactly this respect. The age-of-youngest-child exemption set how young a child could be before its mother faced work requirements, ranging from immediate mandates in the least generous states to multi-year shelters in the most generous. Table 7 lets the treatment effect vary with this threshold, drawn from the Department of Health and Human Services and measured in months, with the full schedule in Table A5 and its geography in Figure A9. The caregiving account predicts that harm should shrink as the exemption lengthens.

[Table 7 About Here]

That is what the estimates show, and they show it for the traits that carry the baseline result. Across all states in Panel A, a longer exemption significantly dampens the agreeableness and neuroticism declines, and the full-index estimate moves in the same direction though it is not precisely estimated. Panel B, which drops states at the federal default and those still phasing in their rules, reproduces the pattern, with the agreeableness moderation slightly larger and again significant. The protection concentrates on agreeableness, the trait most robustly affected in the baseline, so the reform's imprint is measurably lighter in the states that shielded early maternal care the longest.

5.4 When Does Exposure Matter?

My baseline treatment measures the share of the conception-to-age-five window spent under reform, but because reform is permanent, a child exposed early is necessarily exposed later as well, so the estimate pools exposure across the whole of childhood. Resolving *when* in childhood exposure matters is therefore both an interpretive and a substantive question. The developmental logic that motivates this paper makes a clear prediction. If noncognitive skills are laid down through the continuity and quality of care in the earliest years, then the harm should be largest for children exposed from birth or before and should fade as the age at first exposure rises, once the sensitive window has already closed, additional exposure should do little.

I test this by replacing the single dose with a series of indicators for the child's age when reform arrived in their birth state, interacted with target status, and I plot the resulting coefficients in Figure 1. The horizontal axis runs from reform already in place before birth, at the left, through reform arriving in early childhood, to reform arriving only in later childhood, at the right. Because exposure is greatest at the left and declines moving rightward, the developmental prediction is a gradient rising from negative values at early ages toward zero at later ages. The comparison group serves as the internal placebo, and age ten is the omitted reference.

[Fig. 1 About Here]

Figure 1 traces exactly this gradient. The estimated effect on the composite index is most negative for children exposed in utero and through the first years of life, around −0.2 to −0.35 SD, and attenuates steadily toward zero as the age at reform rises, reaching effectively zero for children first exposed in later childhood. The linear fit superimposed on the coefficients summarizes the pattern, a positive slope that carries the effect from its early-life trough up to zero. Two features of the figure support the design as much as the interpretation. The coefficients for reform arriving in later childhood, at the right, are flat and centered on zero, indicating that exposure past the early window leaves no imprint. And the coefficients for reform already in place before birth, at the far left, are similarly stable, which is difficult to reconcile with the effect being generated by cohort trends within states rather than by exposure itself.

5.5 Benchmarking the Effect Size

How large is the estimated decline? A composite effect of 0.23 SD moves a fully exposed target child from roughly the median to the 41st percentile of the noncognitive-skill distribution, and the trait-specific declines, 0.31 SD for agreeableness and 0.44 SD for openness, are larger still. Because NCS are measured with different instruments, at different ages, and after very different early-life shocks, no comparison is exact, but a handful of estimates place the effect in a reasonable range.

The closest comparisons involve the same traits. Akee et al. (2018), studying an unconditional cash transfer to Native American households, find effects on the three Big Five traits they measure, agreeableness, conscientiousness, and neuroticism, on the order of 0.21 to 0.27 SD, close to my own magnitudes but reversed in direction, since an inflow of early-life resources raised these traits while a reform that tightened assistance lowered them. Openness shows the same pattern: García et al. (2023) find that the Perry Preschool Project raised it by 0.32 SD, a same-trait, opposite-sign counterpart to my 0.44 SD decline.

Two of these estimates also speak to persistence, which matters because I measure skills in young adulthood, once personality has consolidated. The Perry openness effect is observed at age 54, and Attanasio et al. (2024) find that kangaroo mother care reduces externalizing behavior by 0.18 SD fully twenty years after treatment. That effects of this magnitude remain detectable decades later supports reading my estimate as a lasting imprint on the adult trait rather than a transient childhood change.

The direction of my result matches the smaller set of shocks that, like welfare reform, displaced early parental care. Baker et al. (2019) find that expanded subsidized childcare in Quebec worsened children's socioemotional outcomes by 0.13 to 0.27 SD, and Morando and Platt (2022) find that center-based care in infancy raises externalizing behavior by about 0.11 SD. Both fall in a comparable range and point the same way, reinforcing that early shifts away from parental care can depress socioemotional development.

Across these cash-transfer, preschool, and childcare shocks, effects on the order of 0.2 to 0.4 SD are typical whether early-life resources are added or withdrawn, and they persist into adulthood. My estimate sits squarely within that range. What sets this paper apart is not the size of the effect but its setting and sign, no prior work has estimated welfare reform's effect on noncognitive skills in adulthood, and the reform lowered rather than raised the traits at issue.

5.6 Robustness Checks

In this section, I subject the baseline estimate to the battery of checks described in Section 4. I begin with the two most central to the design, the parallel-trends assumption and the exogeneity of reform timing, and then turn to selective migration, negative weighting, and two checks on unobserved confounding.

The identifying assumption requires that, absent reform, the target and comparison groups would have followed similar trajectories across birth cohorts. I assess this with the event-study specification of Section 4, which replaces the single dose with a series of coefficients by year relative to state reform adoption, omitting the year before adoption as the reference. Figure 2 plots these estimates for the composite index. The pre-adoption coefficients are small and statistically indistinguishable from zero, with a joint test yielding $p = 0.166$, so I find no evidence of differential trends before reform took effect. The effect emerges only from adoption onward; the pattern expected of a treatment that begins with exposure rather than a pre-existing divergence. Because every state in the sample adopts reform within a narrow window and none remains untreated, I confirm this in the stacked event study of Figure A7, in which each adoption cohort is matched to not-yet-treated states and the sub-experiments are stacked with cohort-specific fixed effects. The stacked estimates reproduce the same flat pre-period, with a joint pre-adoption test of $p = 0.275$, so the result does not depend on the comparisons that generate negative weights under staggered adoption.

[Figure 2 About Here]

A second concern is that states timed reform in response to economic or social conditions that also bore on children's development. Table 11 examines this by regressing the timing of state reform adoption, measured as months from January 1992 to the earlier of a state's waiver or TANF date, on a vector of pre-reform state characteristics, across the full universe of reform states and in specifications that add region fixed effects, weight by population, and exclude influential states. In the baseline column, no individual characteristic is a significant predictor of adoption timing, and the pre-reform characteristics jointly explain little of the variation in timing, with an R-squared of 0.109. This weak fit is itself reassuring for the design, echoing the finding of Ziliak et al. (2000) that reform timing was largely unrelated to states' pre-reform economic conditions. The pattern holds across the remaining specifications, where no characteristic emerges as a consistent predictor. The absence of systematic determinants supports the assumption that reform timing is plausibly exogenous to the state conditions relevant for child development, and these characteristics enter the main specification interacted with birth cohort, absorbing any residual correlation directly.

[Table 11 About Here]

Because exposure is assigned by state of birth, families who moved during the exposure window could break the link between assigned and realized policy exposure. Table 12 addresses this by restricting the sample to children who remained in their birth state through early childhood, for whom the assigned exposure reflects the policy environment they actually experienced, and building the specification up sequentially from fixed effects through the full control set. The interaction is stable throughout, at -0.29 in the preferred column against -0.23 in the full sample, so restricting to children whose early-life exposure is measured without migration error, if anything, strengthens the estimate, and selective migration does not account for the effect.

[Table 12 About Here]

Next, since states adopt reform at different times and the effect may vary across cohorts and states, the standard two-way fixed-effects estimator can place negative weights on some comparisons, allowing already-treated cohorts to serve as controls for later-treated ones. I gauge the severity of this in Table 8, decomposing the estimand into its 104-underlying group-by-period effects following de Chaisemartin and d'Haultfoeuille (2020). Only 33 receive negative weights, and these sum to −0.10, about a tenth of the total weight mass, while the sensitivity measures indicate that the estimate would change sign only under treatment-effect heterogeneity far larger than is plausible here. I then estimate the stacked design of Table 9, in which each adoption cohort is matched to not-yet-treated states and the sub-experiments are stacked with cohort-specific fixed effects, so that identification never draws on forbidden comparisons. The stacked composite estimate is −0.21 and significant, close to the baseline, and the subscale pattern is preserved, with agreeableness at −0.26 and openness at −0.62, both significant. That the effect survives a design purged of negative-weight comparisons indicates it is not an artifact of the staggered rollout.

[Table 9 About Here]

Also, because the exogeneity of timing cannot be established conclusively through covariate balance alone, I complement the timing regression with a non-parametric randomization test that probes directly whether the baseline estimate could arise by chance, in the spirit of Ferrara et al. (2012). The logic is straightforward. If treatment eligibility carries a real signal, then randomly reassigning target status across individuals should sever any systematic relationship between early-life exposure and young-adult NCS. For each permutation r, I draw a placebo target indicator $T_i^{(r)}$, form the corresponding placebo interaction $RE_{isc}^{\text{IU-5}} \times T_i^{(r)}$, re-estimate the baseline specification, and store the coefficient $\widehat{\beta}^{(r)}$. Repeating this 1,000 times yields an empirical null distribution $\left\{\widehat{\beta}^{(r)}\right\}_{r=1}^{R}$ under the hypothesis of no effect. The randomization p-value is the share of placebo estimates at least as large in magnitude as the observed coefficient,

$$p = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\left(\left|\widehat{\beta}^{(r)}\right| \geq \left|\widehat{\beta}\right|\right).$$

Figure 3 plots the resulting distribution against the observed estimate. The placebo coefficients center on zero, as expected when eligibility is assigned at random, and the observed effect of −0.23 lies in the extreme left tail, yielding an empirical p-value of 0.002. Because the test is distribution-free, it does not rely on the asymptotic approximation behind the clustered standard errors, and together with the timing regression it provides a strong final check on identification, the one evaluating whether reform timing is predictable, the other whether the observed effect exceeds what arbitrary assignment of eligibility would produce.

[Figure 3 About Here]

Finally, I bound the scope for omitted-variable bias following Oster (2019). Table 13 reports, for each outcome, the coefficient's movement as controls are added and the degree of selection on unobservables, relative to selection on the included controls, that would be required to drive the estimate to zero. For the composite index the coefficient is essentially unchanged as controls enter, moving from −0.217 to −0.226, and the implied δ is 5.83, meaning unobservables would need to be more than five times as important as the full set of observed controls to explain away the effect. Agreeableness is similarly stable with a δ of 1.72. The identified set excludes zero for the composite and for agreeableness, so both the coefficient stability and the Oster bounds point to an effect that is robust to plausible unobserved confounding.

[Table 13 About Here]

5.7 Labor-Market Returns to NCS

The estimates so far show that welfare reform lowered NCS, with the decline concentrated in agreeableness and openness. I next ask whether these socioemotional changes are reflected in adult economic outcomes. I do so in two steps. First, I estimate the association between NCS and adult labor-market outcomes in an augmented Mincer framework. Second, I estimate the reduced-form effect of early-life reform exposure on adult earnings and employment.

Table 10 reports the returns to skill. Columns 1 and 4 present the baseline human-capital specification, while columns 2 and 5 add the composite NCS index. The composite index is not significantly related to earnings, with a coefficient of −0.04. Taken alone, this could suggest that NCS is not priced in the labor market. The disaggregated estimates show otherwise. When the five subscales are entered separately, the composite null masks offsetting associations across traits. Agreeableness is associated with lower earnings, with a coefficient of −0.21, and openness is also negatively associated with earnings, at −0.11.

By contrast, conscientiousness is positively associated with earnings, and neuroticism has a smaller positive association. The composite therefore averages together traits with different labor-market prices. For employment, the pattern is similar. Employment is more strongly related to conscientiousness and neuroticism, and is not meaningfully related to agreeableness or openness.

This pricing pattern helps interpret the reduced-form labor-market estimates. Welfare reform reduced agreeableness and openness, the two traits that are not positively rewarded in the earnings regressions. A mechanical decomposition that combines the estimated skill changes with these descriptive returns would therefore not predict an earnings loss and may even imply a gain. The reduced-form estimates, however, show neither a gain nor a loss. Table A3 reports no detectable effect of early-life reform exposure on adult earnings or employment.

The absence of labor-market effects does not imply that the NCS losses are unimportant. Rather, it shows that the traits affected by the reform are not the traits most strongly rewarded in earnings or employment. The developmental cost of welfare reform therefore appears on socioemotional margins that standard labor-market outcomes do not capture.

6 Discussion

The results show that welfare reform had lasting effects on the socioemotional development of the next generation. Early-life exposure lowered NCS in young adulthood, with the decline concentrated in agreeableness and openness. This finding extends the second-generation welfare reform literature by showing that the reform's long-run effects were not limited to schooling, fertility, family formation, or adolescent behavior. The reform also shaped a dimension of human capital that is less visible in conventional evaluations but central to how individuals regulate behavior, relate to others, and navigate adult roles.

The pattern of results points to early caregiving conditions as the most plausible pathway. The effects are largest for children exposed in utero and during the first years of life, and they fade as the age at first exposure rises. This timing is consistent with models of skill formation in which early environments shape later capabilities through self-productivity and dynamic complementarity (Cunha and Heckman, 2007; Heckman et al., 2013). It is also consistent with developmental evidence emphasizing the importance of stable, responsive care in early childhood (Phillips and Shonkoff, 2000). The mechanism estimates reinforce this interpretation. I find little evidence that reform exposure raised household income or maternal employment, but stronger evidence that it changed the early caregiving environment through formal care arrangements and household composition. These estimates are suggestive, since the mechanism variables are measured at a point in time, but they point away from a simple income story and toward changes in the conditions under which early care was provided.

The AYCE evidence provides further support for this interpretation. If work requirements harmed socioemotional development by displacing early maternal care, then the harm should be weaker in states that exempted mothers with very young children for longer periods. That is what the estimates indicate. Longer exemptions dampen the declines in the traits most affected in the baseline results, especially agreeableness. This evidence complements Herbst (2017), who shows that the AYCE is relevant for children's early developmental outcomes. My results extend that logic into young adulthood. The point is not that maternal employment is inherently harmful. Rather, work requirements imposed during the earliest caregiving window may carry developmental costs when they change care arrangements without generating offsetting gains in household resources.

The heterogeneity results should be read cautiously but are informative. The effects are most clearly estimated among girls, although the interaction tests do not support a strong claim that girls were affected differently from boys. I therefore interpret the gender pattern as evidence on where the effect is most detectable, not as a clean gender contrast. Still, it fits a broader literature showing that boys and girls may respond differently to early family environments and disadvantage (Autor et al., 2019). A similar caution applies to race. The effects are more visible among White children, but subgroup differences are imprecisely estimated. This pattern may reflect differences in policy environments, program administration, baseline resources, or statistical precision rather than a structural racial difference in treatment effects. This interpretation is consistent with evidence that welfare implementation and its consequences varied across racial and administrative contexts (Soss et al., 2011).

The labor-market results clarify what kind of cost the reform generated. The traits reduced by welfare reform, agreeableness and openness, are not the traits most positively rewarded in earnings or employment. In the descriptive returns estimates, conscientiousness and emotional stability are more strongly associated with favorable labor-market outcomes, while agreeableness and openness are weakly or negatively priced. This pattern is consistent with evidence that labor-market returns vary sharply across personality traits, including work showing limited or negative earnings returns to agreeableness (Flinn et al., 2025). It therefore is not surprising that the reduced-form estimates show no detectable effects on adult earnings or employment. The absence of labor-market effects does not imply the NCS losses are unimportant. It means the affected traits matter along margins that earnings and employment do not fully price.

This distinction is central for policy evaluation. Welfare reform is often assessed through caseloads, maternal employment, income, schooling, and adult attainment. Those outcomes remain important, but they do not exhaust the ways safety-net policy can shape children's development. A reform can leave persistent socioemotional effects even when conventional economic outcomes show little change. The findings therefore suggest that evaluations of safety-net reform should attend not only to benefit generosity and work incentives, but also to the timing and caregiving conditions under which those incentives operate. For children exposed in early life, the design of the safety net may matter as much for the quality and continuity of care as for household income.

6.1 Conclusion

This paper examines whether early-life exposure to welfare reform shaped the noncognitive skills of the next generation. The 1990s restructuring of cash assistance changed the incentives and constraints facing low-income mothers, but its implications for children were theoretically ambiguous. Increased employment and stronger work incentives could have improved household resources and routines, while time limits, sanctions, and work requirements could have increased stress, reduced parental time, and disrupted early care. Nearly three decades later, the children exposed to these reforms can now be observed as young adults, making it possible to assess whether the reform left persistent marks on socioemotional development.

Using the PSID Child Development Supplement and Transition to Adulthood Supplement, I estimate a triple-difference design that exploits variation in welfare reform timing across states, birth cohorts, and maternal likelihood of being affected by the reform.

I find that exposure from conception through age five lowered noncognitive skills in young adulthood by about 0.23 standard deviations among individuals born to likely affected mothers. The decline is concentrated in agreeableness and openness, and the age-at-exposure pattern indicates that the effects are largest when exposure occurs during the earliest years of life. Mechanism analyses point more strongly to changes in the early caregiving environment than to household income or maternal employment.

The results suggest that welfare reform affected the next generation along a margin of human capital that has received limited attention. Prior evaluations have focused largely on caseloads, maternal employment, income, schooling, fertility, family formation, and other conventional outcomes. These outcomes are important, but they may miss persistent changes in socioemotional development. Consistent with this interpretation, I find no detectable effect of early-life exposure on adult earnings or employment, in part because the traits affected by the reform are not the traits most strongly rewarded in the labor market.

The policy implication is not that work-oriented safety-net reform is necessarily harmful to children. Rather, the findings show that the design and timing of such reforms matter. Work requirements and sanctions imposed during the earliest years of childhood may alter caregiving conditions in ways that persist into adulthood, even when conventional economic outcomes show little change. Evaluations of safety-net policy should therefore account not only for employment, income, and program participation, but also for the developmental environments created for children growing up under these reforms.

References

Akee, R., Copeland, W., Costello, E. J., and Simeonova, E. (2018). How does household income affect child personality traits and behaviors? *American Economic Review*, 108(3):775–827.

Aughinbaugh, A. and Gittleman, M. (2004). Maternal employment and adolescent risky behavior. *Journal of Health Economics*, 23(4):815–838.

Autor, D., Figlio, D., Karbownik, K., Roth, J., and Wasserman, M. (2019). Family disadvantage and the gender gap in behavioral and educational outcomes. *American Economic Journal: Applied Economics*, 11(3):338–381.

Baker, M., Gruber, J., and Milligan, K. (2019). The long-run impacts of a universal child care program. *American Economic Journal: Economic Policy*, 11(3):1–26.

Bastian, J., Bian, L., and Grogger, J. (2021a). How did safety-net reform affect early adulthood among adolescents from low-income families? *National Tax Journal*, 74(3):825–865.

Bastian, J., Bian, L., and Grogger, J. (2021b). How did safety-net reform affect early adulthood among adolescents from low-income families? *National Tax Journal*, 74(3):825–865.

Benjamini, Y., Krieger, A. M., and Yekutieli, D. (2006). Adaptive linear step-up procedures that control the false discovery rate. *Biometrika*, 93(3):491–507.

Bennhoff, F. H., García, J. L., and Leaf, D. E. (2024). The dynastic benefits of early-childhood education: Participant benefits and family spillovers. *Journal of Human Capital*, 18(1):44–73.

Bitler, M. and Hoynes, H. W. (2010). The state of the safety net in the post-welfare reform era. NBER Working Paper w16504, National Bureau of Economic Research.

Blank, R. M. (2002). Evaluating welfare reform in the united states. *Journal of Economic Literature*, 40(4):1105–1166.

Blank, R. M. (2009). What we know, what we don't know, and what we need to know about welfare reform. In *Welfare Reform and Its Long-Term Consequences for America's Poor*, pages 22–58. Cambridge University Press.

Bleidorn, W., Schwaba, T., Zheng, A., Hopwood, C. J., Sosa, S. S., Roberts, B. W., and Briley, D. A. (2022). Personality stability and change: A meta-analysis of longitudinal studies. *Psychological Bulletin*, 148(7-8):588.

Bolbocean, C. and Tylavsky, F. A. (2021). The impact of safety net programs on early-life developmental outcomes. *Food Policy*, 100:102018.

Chase-Lansdale, P. L., Cherlin, A. J., Guttmannova, K., Fomby, P., Ribar, D. C., and Coley, R. L. (2011). Long-term implications of welfare reform for the development of adolescents and young adults. *Children and Youth Services Review*, 33(5):678–688.

Chyi, H. and Ozturk, O. D. (2013). The effects of single mothers' welfare use and employment decisions on children's cognitive development. *Economic Inquiry*, 51(1):675–706.

Chyi, H., Ozturk, O. D., and Zhang, W. (2014). Welfare reform and children's early cognitive development. *Contemporary Economic Policy*, 32(4):729–751.

Corman, H., Dave, D., Kalil, A., and Reichman, N. E. (2017). Effects of maternal work incentives on youth crime. *Labour Economics*, 49:128–144.

Corman, H., Dave, D. M., Schwartz-Soicher, O., and Reichman, N. E. (2022). Effects of welfare reform on household food insecurity across generations. *Economics & Human Biology*, 45:101101.

Crouse, G. (1999). State implementation of major changes to welfare policies, 1992–1998. Technical report, U.S. Department of Health and Human Services, Washington, DC.

Cunha, F. and Heckman, J. (2007). The technology of skill formation. *American Economic Review*, 97(2):31–47.

Currie, J. and Almond, D. (2011). Human capital development before age five. In *Handbook of Labor Economics*, volume 4, pages 1315–1486. Elsevier.

Dave, D., Corman, H., Kalil, A., Schwartz-Soicher, O., and Reichman, N. E. (2021). Intergenerational effects of welfare reform: Adolescent delinquent and risky behaviors. *Economic Inquiry*, 59(1):199–216.

Dave, D. M., Corman, H., and Reichman, N. E. (2012). Effects of welfare reform on education acquisition of adult women. *Journal of Labor Research*, 33(2):251–282.

de Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.

Dore, E. C., Hamad, R., Komro, K. A., and Livingston, III, M. D. (2025). The long-term health effects of welfare reform. *Social Science & Medicine*, 371:117878.

Duncan, G. J., Huston, A. C., and Weisner, T. S. (2007). *Higher Ground: New Hope for the Working Poor and Their Children*. Russell Sage Foundation.

Dunifon, R., Kalil, A., and Danziger, S. K. (2003). Maternal work behavior under welfare reform: How does the transition from welfare to work affect child development? *Children and Youth Services Review*, 25(1-2):55–82.

Esping-Andersen, G. and Cimentada, J. (2018). Ability and mobility: The relative influence of skills and social origin on social mobility. *Social Science Research*, 75:13–31.

Fletcher, J. M. and Wolfe, B. (2016). The importance of family income in the formation and evolution of non-cognitive skills in childhood. *Economics of Education Review*, 54:143–154.

Flinn, C. J., Todd, P. E., and Zhang, W. (2025). Labor market returns to personality: A job search approach to understanding gender gaps. *Journal of Political Economy*, 133(4):1169–1234.

García, J. L., Heckman, J. J., and Ronda, V. (2023). The lasting effects of early-childhood education on promoting the skills and social mobility of disadvantaged african americans and their children. *Journal of Political Economy*, 131(6):1477–1506.

Gennetian, L. A., Duncan, G., Knox, V., Vargas, W., Clark-Kauffman, E., and London, A. S. (2004). How welfare policies affect adolescents' school outcomes: A synthesis of evidence from experimental studies. *Journal of Research on Adolescence*, 14(4):399–423.

Gennetian, L. A. and Miller, C. (2002). Children and welfare reform: A view from an experimental welfare program in minnesota. *Child Development*, 73(2):601–620.

Goldberg, L. R. (1993). The structure of phenotypic personality traits. *American Psychologist*, 48(1):26.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

Grogger, J. (2003). The effects of time limits, the eitc, and other policy changes on welfare use, work, and income among female-headed families. *Review of Economics and Statistics*, 85(2):394–408.

Grogger, J. T. and Karoly, L. A. (2005). *Welfare Reform: Effects of a Decade of Change*. Harvard University Press.

Hartley, R. P., Lamarche, C., and Ziliak, J. P. (2022). Welfare reform and the intergenerational transmission of dependence. *Journal of Political Economy*, 130(3):523–565.

Heckman, J. J. (2000). Policies to foster human capital. *Research in Economics*, 54(1):3–56.

Heckman, J. J. (2008). The case for investing in disadvantaged young children. *CESifo DICE Report*, 6(2):3–8.

Heckman, J. J. and Mosso, S. (2014). The economics of human development and social mobility. *Annual Review of Economics*, 6(1):689–733.

Heflin, C. M. and Acevedo, S. K. (2011). Non-income effects of welfare receipt on early childhood cognitive scores. *Children and Youth Services Review*, 33(5):634–643.

Herbst, C. M. (2017). Are parental welfare work requirements good for disadvantaged children? evidence from age-of-youngest-child exemptions. *Journal of Policy Analysis and Management*, 36(2):327–357.

Hofferth, S. L. (1999). Childcare, maternal employment, and public policy. *The Annals of the American Academy of Political and Social Science*, 563(1):20–38.

Hoynes, H., Schanzenbach, D. W., and Almond, D. (2016). Long-run impacts of childhood access to the safety net. *American Economic Review*, 106(4):903–934.

Humphries, J. E. and Kosse, F. (2017). On the interpretation of non-cognitive skills: What is being measured and why it matters. *Journal of Economic Behavior & Organization*, 136:174–185.

Kaestner, R. and Chan Lee, W. (2005). The effect of welfare reform on prenatal care and birth weight. *Health Economics*, 14(5):497–511.

Kalil, A., Corman, H., Dave, D., Schwarz-Soicher, O., and Reichman, N. E. (2023). Welfare reform and the quality of young children's home environments. *Demography*, 60(6):1791–1813.

Kautz, T., Heckman, J. J., Diris, R., Ter Weel, B., and Borghans, L. (2014). Fostering and measuring skills: Improving cognitive and non-cognitive skills to promote lifetime success. Nber working paper, National Bureau of Economic Research.

Kling, J. R., Liebman, J. B., and Katz, L. F. (2007). Experimental analysis of neighborhood effects. *Econometrica*, 75(1):83–119.

Miller, A. R. and Zhang, L. (2009). The effects of welfare reform on the academic performance of children in low-income households. *Journal of policy Analysis and Management*, 28(4):577–599.

Milligan, K. and Stabile, M. (2011). Do child tax benefits affect the well-being of children? evidence from canadian child benefit expansions. *American Economic Journal: Economic Policy*, 3(3):175–205.

Morris, P., Gennetian, L. A., Duncan, G. J., and Huston, A. (2009). *How Welfare Policies Affect Child and Adolescent School Performance: Investigating Pathways of Influence with Experimental Data*. Cambridge University Press, Cambridge, UK.

Offner, P. (2005). Welfare reform and teenage girls. *Social Science Quarterly*, 86(2):306–322.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.

Phillips, D. A. and Shonkoff, J. P., editors (2000). *From Neurons to Neighborhoods: The Science of Early Childhood Development*. National Academy Press, Washington, DC.

Pilkauskas, N. V., Brooks-Gunn, J., and Waldfogel, J. (2018). Maternal employment stability in early childhood: Links with child behavior and cognitive skills. *Developmental Psychology*, 54(3):410.

Reichman, N. E., Corman, H., Dave, D., Kalil, A., and Schwartz-Soicher, O. (2023). Effects of welfare reform on positive health and social behaviors of adolescents. *Children*, 10(2):260.

Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in stata using boottest. *The Stata Journal*, 19(1):4–60.

Schoeni, R. F. and Blank, R. M. (2000). What has welfare reform accomplished? impacts on welfare participation, employment, income, poverty, and family structure. Working paper, National Bureau of Economic Research.

Soss, J., Fording, R. C., and Schram, S. (2011). *Disciplining the Poor: Neoliberal Paternalism and the Persistent Power of Race*. University of Chicago Press.

UKCPR (2026). UKCPR National Welfare Data, 1980–2023. University of Kentucky Center for Poverty Research, Gatton College of Business and Economics, University of Kentucky, Lexington, KY.

Vaughn, C. (2022). Long-run impact of welfare reform on educational attainment and family structure. Available at SSRN 4131589.

Vaughn, C. N. (2023). Welfare reform and childhood health status and utilization. *Children and Youth Services Review*, 150:107004.

Walker, S. P., Chang, S. M., Wright, A. S., Pinto, R., Heckman, J. J., and Grantham-McGregor, S. M. (2022). Cognitive, psychosocial, and behaviour gains at age 31 years from the jamaica early childhood stimulation trial. *Journal of Child Psychology and Psychiatry*, 63(6):626–635.

Wolfers, J. (2006). Did unilateral divorce laws raise divorce rates? a reconciliation and new results. *American Economic Review*, 96(5):1802–1820.

Ziliak, J. P. (2015). Temporary assistance for needy families. In *Economics of Means-Tested Transfer Programs in the United States, Volume 1*, pages 303–393. University of Chicago Press.

Ziliak, J. P., Figlio, D. N., Davis, E. E., and Connolly, L. S. (2000). Accounting for the decline in afdc caseloads: Welfare reform or the economy? *Journal of Human Resources*, pages 570–586.

Tables and Figures

Table 1: Summary Statistics

	N	Mean	SD
Outcomes			
Noncognitive Skills Index	2837	0.009	0.420
Agreeableness	2837	-0.004	0.713
Conscientiousness	2837	-0.009	0.732
Extraversion	2837	0.016	0.753
Neuroticism	2837	0.042	0.794
Openness	2837	0.001	0.789
Treatment			
Exposure (IU-Age 5)	2837	0.306	0.316
Target	2837	0.153	0.360
Young Adult Controls			
Age at Survey	2837	23.435	2.723
Female	2837	0.487	0.500
Nonwhite	2837	0.266	0.442
Grandparent Present	2837	0.024	0.154
Maternal Controls			
Mother's Age	2837	34.145	7.443
< High School	2837	0.151	0.358
High School	2837	0.312	0.464
Some College	2837	0.460	0.499
Pre-Reform State Level Characteristics			
Unemployment Rate	2837	6.314	1.368
ln(TANF Benefit)	2837	12.483	0.970
ln(SNAP Benefit)	2837	13.104	0.836
Poverty Rate	2837	13.511	3.636
Minimum Wage	2837	3.757	0.560
EITC Rate	2837	0.011	0.067

Notes: Survey-weighted summary statistics for the analytic sample, drawn from the PSID Child Development Supplement and Transition into Adulthood. NCS measures are standardized z-scores.

Table 2: Baseline Estimation

	(1)	(2)	(3)	(4)	(5)
IU-Age 5	-0.031	-0.033	-0.031	-0.048	-0.053
	(0.103)	(0.103)	(0.104)	(0.099)	(0.104)
IU-Age 5 × Target	-0.234***	-0.228***	-0.226***	-0.226***	-0.227***
	(0.083)	(0.080)	(0.078)	(0.081)	(0.082)
Bootstrap p-value	0.012	0.015	0.011	0.014	0.018
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Mother Controls	No	Yes	Yes	Yes	Yes
Young Adult Controls	No	No	Yes	Yes	Yes
State Controls	No	No	No	Yes	Yes
State Linear Trends	No	No	No	No	Yes
Observations	2837	2837	2837	2837	2837
Adj. R-squared	0.047	0.050	0.075	0.085	0.097

Notes: This table reports estimates from a difference-in-difference-in-differences (DDD) specification examining how early-life exposure to welfare reform affects the child's standardized noncognitive skills (NCS) index, a z-score constructed from 15 Big Five personality items measured in young adulthood. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less. IU-Age 5 × Target captures the causal effect of welfare reform for likely-affected children, while the constituent terms represent baseline differences across exposure and maternal-education groups. Column (1) includes only birth-cohort, survey-year, target, and state fixed effects; columns (2)-(4) progressively add maternal characteristics (education, age), young adult controls (age, gender, race, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate). Column (5) adds state-specific linear birth-cohort trends as a robustness check; column (4) is the preferred specification. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. Bootstrap p-values denote wild cluster bootstrap values for the DDD term (9,999 replications, Rademacher weights). * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 3: Big Five Subscales

	(1) Agreeableness	(2) Conscientiousness	(3) Extraversion	(4) Neuroticism	(5) Openness
IU-Age 5	0.007	-0.182	-0.229	-0.164	0.332*
	(0.140)	(0.146)	(0.138)	(0.188)	(0.186)
IU-Age 5 × Target	-0.314**	-0.162	-0.073	-0.140	-0.437**
	(0.130)	(0.153)	(0.172)	(0.147)	(0.192)
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Mother Controls	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes
Unadjusted p	0.021	0.296	0.674	0.346	0.029
BKY q-value	0.044	0.296	0.674	0.346	0.044
Bootstrap p-value	0.030	0.317	0.679	0.397	0.034
Observations	2837	2837	2837	2837	2837

Notes: This table reports difference-in-difference-in-differences (DDD) estimates for each of the five Big Five personality subscales, measured as standardized z-scores in young adulthood. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less. The interaction IU-Age 5 × Target is the causal DDD estimate. All columns include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. To account for testing five outcomes simultaneously, I report multiple-hypothesis-adjusted p-values for the interaction term: BKY denotes the Benjamini-Krieger-Yekutieli (2006) adaptive two-stage q-values, controlling the false discovery rate at q = 0.10. Bootstrap p-values denote wild cluster bootstrap values for the interaction term (9,999 replications, Rademacher weights). Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 4: Gender Heterogeneity — Full Index and Subscales

	(1) Full Index	(2) Agreeableness	(3) Conscientiousness	(4) Extraversion	(5) Neuroticism	(6) Openness
IU-Age 5 (Male)	-0.041	-0.016	-0.185	-0.085	-0.041	-0.016
	(0.100)	(0.161)	(0.129)	(0.194)	(0.100)	(0.161)
IU-Age 5 × Target (Male)	-0.170	-0.249	0.073	-0.005	-0.170	-0.249
	(0.137)	(0.191)	(0.226)	(0.242)	(0.137)	(0.191)
Female	-0.024	0.131**	0.059	0.256***	-0.024	0.131**
	(0.045)	(0.060)	(0.063)	(0.080)	(0.045)	(0.060)
IU-Age 5 × Female	-0.015	0.046	0.004	-0.295	-0.015	0.046
	(0.097)	(0.108)	(0.138)	(0.215)	(0.097)	(0.108)
IU-Age 5 × Target × Female	-0.092	-0.119	-0.398	-0.068	-0.092	-0.119
	(0.162)	(0.224)	(0.288)	(0.223)	(0.162)	(0.224)
Female Total Effect	-0.262	-0.367	-0.326	-0.074	-0.262	-0.367
SE	0.095	0.156	0.208	0.168	0.095	0.156
p-value	0.009	0.024	0.126	0.663	0.009	0.024
Mother Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2837	2837	2837	2837	2837	2837
Adj. R-squared	0.084	0.073	0.045	0.057	0.084	0.073

Notes: This table reports difference-in-difference-in-differences (DDD) estimates allowing the effect of welfare reform to vary by child gender, for the composite NCS index and each Big Five subscale (all standardized). The reference group is male children: "IU-Age 5 × Target (Male)" is the DDD estimate for boys, and the triple interaction "IU-Age 5 × Target × Female" is the additional effect for girls relative to boys. "Female Total Effect" reports the implied DDD estimate for girls (the sum of the male effect and the female differential), with its standard error and p-value obtained by linear combination. All specifications include maternal characteristics (education, age), young adult controls (age, race, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate), together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Race Heterogeneity — Full Index and Subscales

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Index	Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
IU-Age 5 (White)	-0.011	0.046	-0.172	-0.208	-0.136	0.422*
	(0.091)	(0.123)	(0.139)	(0.130)	(0.178)	(0.212)
IU-Age 5 × Target (White)	-0.293**	-0.467*	-0.010	-0.262	-0.115	-0.608**
	(0.128)	(0.255)	(0.201)	(0.256)	(0.223)	(0.288)
Nonwhite	0.175***	0.063	0.063	0.070	0.344***	0.331***
	(0.038)	(0.057)	(0.063)	(0.092)	(0.051)	(0.078)
IU-Age 5 × Nonwhite	-0.173*	-0.234*	-0.029	-0.054	-0.198	-0.345*
	(0.095)	(0.124)	(0.130)	(0.202)	(0.183)	(0.195)
IU-Age 5 × Target × Nonwhite	0.216	0.418	-0.207	0.335	0.091	0.443
	(0.148)	(0.289)	(0.241)	(0.257)	(0.235)	(0.369)
Nonwhite Total Effect	-0.076	-0.049	-0.217	0.073	-0.023	-0.165
SE	0.102	0.147	0.185	0.231	0.182	0.256
p-value	0.458	0.738	0.249	0.753	0.899	0.524
Mother Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2837	2837	2837	2837	2837	2837
Adj. R-squared	0.087	0.078	0.043	0.054	0.218	0.075

Notes: This table reports difference-in-difference-in-differences (DDD) estimates allowing the effect of welfare reform to vary by child race, for the composite NCS index and each Big Five subscale (all standardized). The reference group is White children: "IU-Age 5 × Target (White)" is the DDD estimate for White children, and the triple interaction "IU-Age 5 × Target × Nonwhite" is the additional effect for Nonwhite children relative to White children. "Nonwhite Total Effect" reports the implied DDD estimate for Nonwhite children (the sum of the White effect and the nonwhite differential), with its standard error and p-value obtained by linear combination. All specifications include maternal characteristics (education, age), young adult controls (age, gender, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate), together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Mechanism Analysis

	(1) Informal Childcare	(2) Formal Childcare	(3) Income	(4) Employment	(5) Distress	(6) Young Child	(7) Mother–Child
IU-Age 5	0.083	0.030	3.651	0.121^{*}	-0.313	0.231	-0.038
	(0.052)	(0.045)	(12.588)	(0.063)	(0.817)	(0.440)	(0.070)
IU-Age 5 × Target	-0.055	0.075^{*}	13.592	-0.026	0.845	1.142^{**}	-0.032
	(0.081)	(0.040)	(32.158)	(0.097)	(1.306)	(0.445)	(0.088)
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Maternal controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1863	1863	1868	1868	1144	1868	1863
Adj. R-squared	0.101	0.087	0.085	0.090	0.190	0.643	0.066

Notes: This table reports difference-in-difference-in-differences (DDD) estimates of the effect of early-life welfare-reform exposure on candidate mediating channels measured in the 1997 Child Development Supplement: childcare arrangements (informal, formal), household income, maternal employment, maternal psychological distress, presence of a young child, and mother-child closeness. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less, and IU-Age 5 × Target is the DDD estimate. All models include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 7: Age-of-Youngest-Child Exemption (AYCE)

	(1) Full Index	(2) Agreeableness	(3) Conscientiousness	(4) Extraversion	(5) Neuroticism	(6) Openness
			PANEL A: All States			
IU-Age 5 × Target	-0.233**	-0.664***	0.049	-0.082	-0.360***	-0.098
	(0.096)	(0.113)	(0.166)	(0.263)	(0.126)	(0.149)
IU-Age 5 × Target × Exemption	0.001	0.011***	-0.010	-0.001	0.006*	-0.001
	(0.003)	(0.003)	(0.006)	(0.008)	(0.004)	(0.005)
Observations	2837	2837	2837	2837	2837	2837
Adj. R-squared	0.061	0.054	0.038	0.039	0.154	0.059
			PANEL B: Phase-in States Excluded			
IU-Age 5 × Target	-0.244**	-0.684***	0.069	0.016	-0.418***	-0.192
	(0.105)	(0.123)	(0.179)	(0.288)	(0.147)	(0.159)
IU-Age 5 × Target × Exemption	0.002	0.012***	-0.011	-0.004	0.008*	0.002
Observations	2200	2200	2200	2200	2200	2200
Adj. R-squared	0.064	0.058	0.038	0.047	0.165	0.057
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Mother Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines whether the effect of welfare reform varies with the generosity of a state's age-of-youngest-child work exemption, the age (in months) below which mothers of young children were exempt from AFDC-JOBS work requirements during the waiver era, drawn from the US Department of Health and Human Services. "IU-Age 5 × Target" is the DDD estimate, and "IU-Age 5 × Target × Exemption" is its interaction with the exemption threshold; a positive interaction indicates that more protective (higher-threshold) states experienced smaller reform-induced harm. Panel A includes all states; Panel B excludes states coded at the federal default and those phasing in their requirements. All models include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

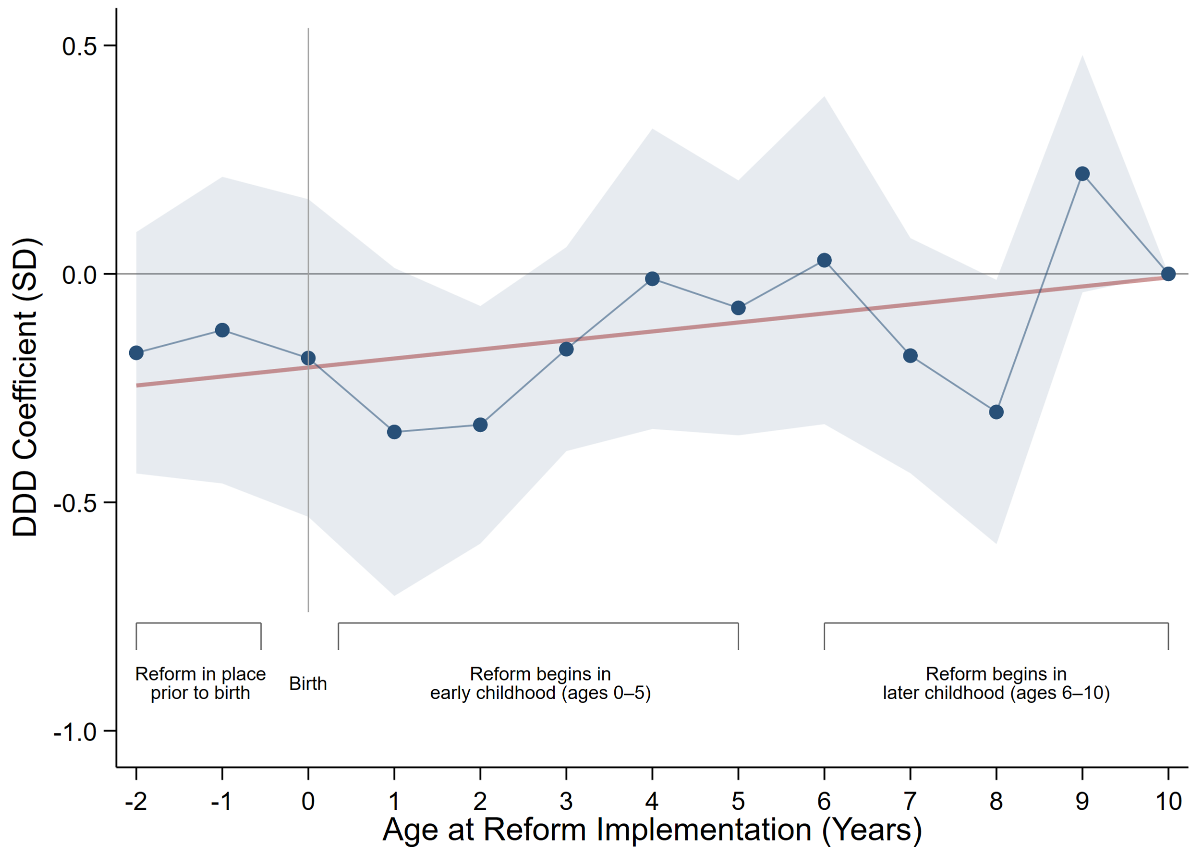

Fig. 1: Effect of Welfare Reform by Age at Exposure
Notes: This figure plots DDD coefficients by the child's age at reform implementation, interacting age-at-reform indicators with target status (age 10 omitted). Because exposure declines from left to right, negative coefficients at younger ages indicate that harm is concentrated among children exposed to reform earliest in life, attenuating toward zero as age at reform rises; the red line is a linear fit summarizing this gradient. The comparison group serves as the internal placebo. The pattern is consistent with a critical-period mechanism in which early-childhood exposure is most consequential. The model includes maternal, young adult, and pre-reform state economic controls, together with survey-year and state fixed effects. Estimates are survey weighted, and shaded bands are 90% confidence intervals from standard errors clustered at the state level.

Table 8: TWFE Weight Decomposition and Sensitivity

Treatment variable: totaltreat05	**ATTs**	**Σ weights**
Positive weights	71	1.1035
Negative weights	33	-0.1035
Total	**104**	**1.0000**

Notes: The two-way fixed-effects estimand is a weighted sum of 104 group-by-period treatment effects; 33 receive negative weights summing to -0.10, roughly a tenth of the total weight mass. The minimum standard deviation of treatment effects required for the estimand and the average treatment effect on the treated to differ in sign is 0.52, and 0.59 for the estimand to have the opposite sign from the effect in every treated cell.

Table 9: Stacked Triple-Difference Estimates: NCS Composite and Big Five Subscales

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Index	Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
IU-Age 5	-0.073	-0.054	-0.027	-0.250	-0.261*	0.229
	(0.087)	(0.123)	(0.154)	(0.192)	(0.141)	(0.228)
IU-Age 5 × Target	-0.213**	-0.264*	-0.238	0.075	-0.023	-0.620***
	(0.104)	(0.141)	(0.170)	(0.250)	(0.195)	(0.230)
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Survey-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Target FE	Yes	Yes	Yes	Yes	Yes	Yes
Maternal Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10143	10143	10143	10143	10143	10143
Adj. R-squared	0.090	0.078	0.067	0.064	0.224	0.095

Notes: This table reports stacked triple-difference (DDD) estimates of early-life welfare reform exposure on young-adult noncognitive skills, for the composite index and each of the five Big Five subscales. The stacked design forms one sub-experiment per reform-adoption cohort (1992 through 1996). Within each sub-experiment, treated units are children in states adopting reform in the cohort year, and control units are drawn from not-yet-treated states, those adopting strictly later; because every state adopts reform by 1997, no never-treated group exists, so not-yet-treated states serve as the clean controls. This construction confines all comparisons to within a sub-experiment and rules out the forbidden comparisons between earlier- and later-treated units that can generate negative weights under staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfoeuille, 2020). Each sub-experiment is assigned its own state, birth-cohort, survey-year, and target fixed effects, so children near a cohort boundary may enter multiple sub-experiments under distinct references. The coefficient on IU-Age 5 x Target is the DDD effect of a marginal increase in the share of the in-utero to age-five window spent under reform, for children of low-educated single mothers relative to the comparison group. All outcomes are survey weighted using TAS weights, and standard errors, clustered on state, appear in parentheses. NCS measures are standardized z-scores. *** p<0.01, ** p<0.05, * p<0.10.

Table 10: Returns to Skills: Earnings and Employment

	Earnings (PPML)			Employment (LPM)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Full Index	Subscales	Baseline	Full Index	Subscales
Years of Schooling	0.201***	0.201***	0.202***	0.045***	0.044***	0.043***
	(0.031)	(0.031)	(0.030)	(0.009)	(0.009)	(0.009)
Age	1.033***	1.031***	1.073***	0.064	0.073	0.070
	(0.372)	(0.372)	(0.357)	(0.104)	(0.099)	(0.099)
Age squared	-0.021***	-0.021***	-0.021***	-0.001	-0.001	-0.001
	(0.007)	(0.007)	(0.007)	(0.002)	(0.002)	(0.002)
Female = 1	-0.319***	-0.319***	-0.273***	-0.043**	-0.039*	-0.038
	(0.084)	(0.085)	(0.089)	(0.020)	(0.020)	(0.023)
NCS Index (Full)		-0.036			0.118***	
		(0.112)			(0.031)	
Agreeableness			-0.209***			0.024
			(0.059)			(0.015)
Conscientiousness			0.136**			0.037**
			(0.063)			(0.016)
Extraversion			0.025			0.019
			(0.052)			(0.019)
Neuroticism			0.094*			0.026*
			(0.053)			(0.015)
Openness			-0.108**			0.011
			(0.045)			(0.017)
Observations	1853	1853	1853	1854	1854	1854

Notes: This table reports the estimated labor-market returns to skills in adulthood. Columns (1)-(3) model earnings using Poisson pseudo-maximum-likelihood (PPML) on raw earnings; columns (4)-(6) model employment using a linear probability model. Within each outcome, the baseline columns include only schooling and demographics, the second columns add the full standardized NCS index, and the third columns replace it with the five Big Five subscales. All specifications include state, birth-cohort, and survey-year fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Determinants of State Welfare Reform Timing: Robustness Checks

	(1)	(2)	(3)	(4)	(5)
Population	10.534	9.034	19.432	21.156*	36.143
	(11.846)	(12.386)	(12.027)	(11.409)	(23.311)
Unemployment Rate	0.861	0.566	2.505	2.313	4.572
	(2.225)	(2.338)	(2.426)	(2.324)	(4.248)
TANF Receipts	-6.131	-7.398	-10.969	-4.679	-15.429
	(9.456)	(10.094)	(9.664)	(10.455)	(11.171)
SNAP Receipts	-7.170	-4.342	-10.875	-19.285	-21.635
	(14.347)	(15.442)	(15.414)	(15.490)	(21.358)
Poverty Rate	1.092	1.009	1.310	1.439	2.938
	(1.303)	(1.387)	(1.159)	(1.074)	(1.853)
State Minimum Wage	-2.855	-2.818	-6.950	-5.663	-10.080
	(3.947)	(4.161)	(4.734)	(4.629)	(6.175)
State EITC Rate	43.403	43.848	48.933	43.371	96.977
	(33.881)	(33.015)	(37.684)	(38.534)	(95.825)
Region (Ref: Northeast)					
Midwest			-15.242	-9.844	-20.373*
			(9.892)	(9.993)	(11.170)
South			-10.019	-2.092	
			(10.416)	(10.684)	
West			-16.882	-9.120	-24.918*
			(10.262)	(11.270)	(13.662)
Constant	38.671	43.843	18.198	12.666	-66.722
	(64.283)	(66.828)	(59.490)	(56.304)	(123.946)
Observations	50	50	50	48	34
R-squared	0.109	0.108	0.198	0.178	0.279

Notes: This table regresses the timing of state welfare-reform adoption (months from January 1992 to the earlier of a state's waiver or TANF implementation date) on pre-reform state characteristics, to assess whether adoption timing is systematically related to observable economic and demographic conditions. Column (1) is the baseline; column (2) weights by population; column (3) adds Census region fixed effects; column (4) excludes California and New York; column (5) excludes Southern states. Population, TANF receipts, and SNAP receipts enter in logs. The absence of consistently significant predictors supports the identifying assumption that reform timing is plausibly exogenous to state conditions. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

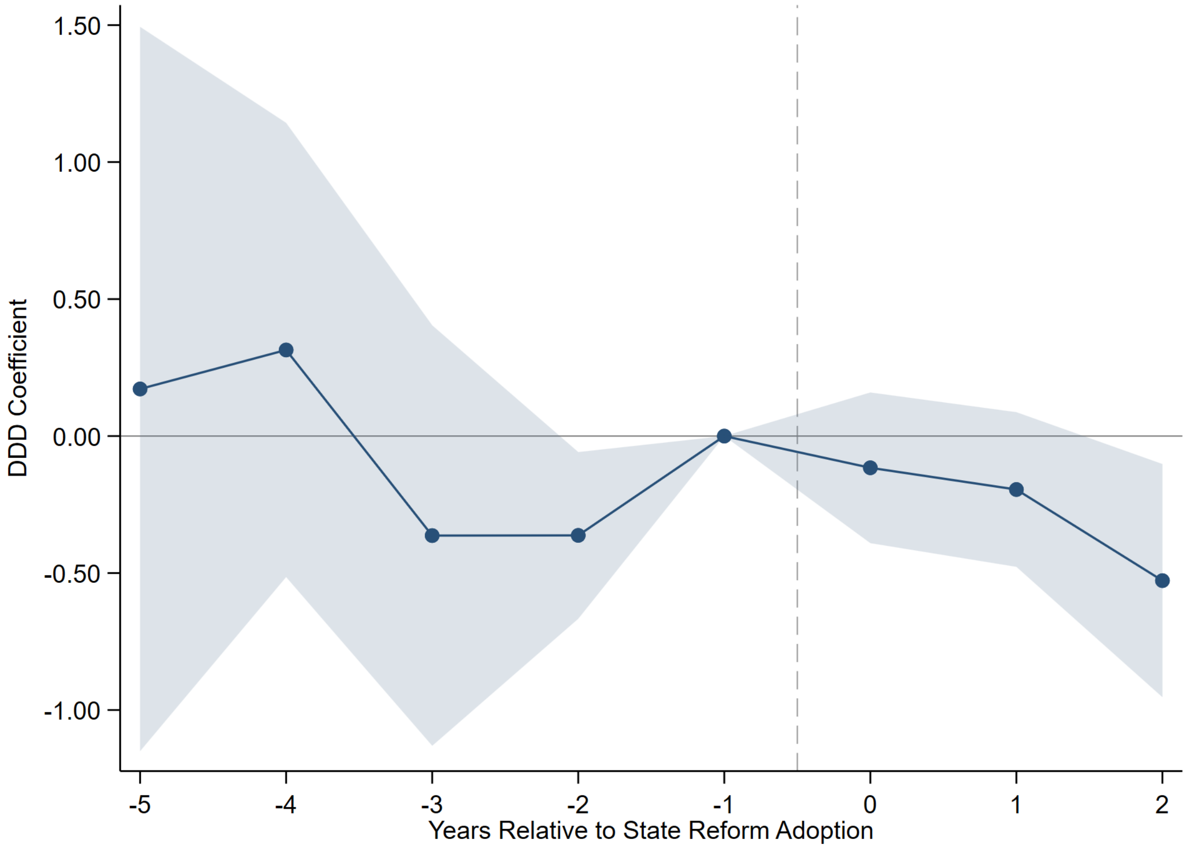

Fig. 2: Event-Study Estimates by Year Relative to State Reform Adoption
Notes: This figure plots the dynamic DDD coefficients from an event-study specification, showing the effect of welfare-reform exposure on the composite NCS index by year relative to state reform adoption. Each point is the interaction of exposure and target with an indicator for a given event year, relative to the omitted reference year ($t = -1$); post-adoption years at $t \geq 2$ are pooled to address thin late-cohort cells. Flat, insignificant pre-period coefficients support the parallel-trends assumption; the joint test of the pre-adoption coefficients yields $p = 0.166$. The model includes maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted, and shaded bands are 90% confidence intervals from standard errors clustered at the state level.

Table 12: Migration Robustness — Stayers Only (Composite NCS Index)

	(1)	(2)	(3)	(4)	(5)
IU-Age 5	-0.081	-0.076	-0.056	-0.063	-0.049
	(0.128)	(0.129)	(0.137)	(0.140)	(0.181)
IU-Age 5 × Target	-0.294**	-0.293***	-0.282**	-0.288**	-0.268**
	(0.110)	(0.101)	(0.104)	(0.112)	(0.119)
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Mother Controls	No	Yes	Yes	Yes	Yes
Young Adult Controls	No	No	Yes	Yes	Yes
State Controls	No	No	No	Yes	Yes
State Linear Trends	No	No	No	No	Yes
Observations	1828	1828	1828	1828	1828
Adj. R-squared	0.074	0.082	0.110	0.108	0.122

Notes: This table restricts the sample to stayers, children whose state of birth equals their exposure state, for whom early-life reform exposure is measured without migration error. Columns add controls progressively: column (1) includes only birth-cohort, survey-year, target, and state fixed effects; columns (2)-(4) add maternal characteristics (education, age), young adult controls (age, gender, race, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate); column (5) adds state-specific linear birth-cohort trends. Column (4) is the preferred specification, and the sample is held fixed across columns. The stability of the IU-Age 5 × Target estimate within the stayer sample indicates that the main results are not driven by selective migration into or out of reform states. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Bounds Estimates for Coefficient Stability and Robustness to Omitted Variable Bias

Treatment variable	**(1)** Baseline coeff. β (SE) $[R^2_U]$	**(2)** Controlled coeff. β (SE) $[R^2_C]$	**(3)** Identified set $[\beta,\ \beta^*(\min\{1, 1.3R^2_C\}, \delta = 1)]$	**(4)** δ for $\beta = 0$ given $R^2_{\max}$
Panel A: Composite NCS Index				
IU-Age 5 × Target	-0.217	-0.226	$[-0.226, -0.326]$	5.827
(SE)	(0.087)	(0.081)		
$[R^2]$	[0.059]	[0.106]		
Panel B: Agreeableness				
IU-Age 5 × Target	-0.317	-0.314	$[-0.314, -0.246]$	1.723
(SE)	(0.119)	(0.130)		
$[R^2]$	[0.065]	[0.095]		
Panel C: Conscientiousness				
IU-Age 5 × Target	-0.141	-0.162	$[-0.162, -0.035]$	1.143
(SE)	(0.146)	(0.153)		
$[R^2]$	[0.046]	[0.066]		
Panel D: Extraversion				
IU-Age 5 × Target	-0.027	-0.073	$[-0.073, -0.109]$	8.629
(SE)	(0.162)	(0.172)		
$[R^2]$	[0.051]	[0.076]		
Panel E: Neuroticism				
IU-Age 5 × Target	-0.283	-0.140	$[-0.140, -0.223]$	16.841
(SE)	(0.169)	(0.147)		
$[R^2]$	[0.099]	[0.235]		
Panel F: Openness				
IU-Age 5 × Target	-0.313	-0.437	$[-0.437, -0.822]$	-2.254
(SE)	(0.183)	(0.192)		
$[R^2]$	[0.067]	[0.098]		

Notes: Each panel reports results for one outcome. Column (1) is the uncontrolled coefficient (fixed effects only); column (2) adds the full control set. The identified set spans the controlled coefficient and the bias-adjusted coefficient beta*(delta=1), with R-squared max set to 1.3 times the controlled R-squared (Oster 2019). Column (4) reports the delta that would shrink the coefficient to zero; delta greater than one (or negative) indicates the estimate is unlikely to be driven by unobservable (Oster 2019; Graham et al. 2017). Estimates are survey weighted with standard errors clustered on state.

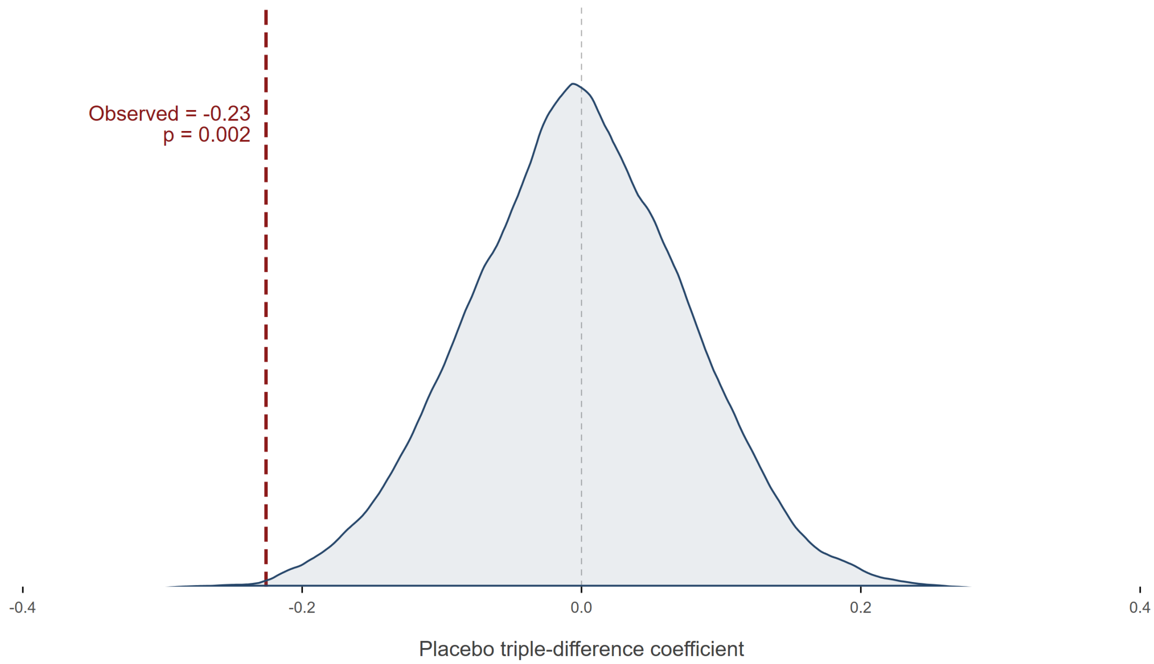

Fig. 3: Randomization Inference: Placebo Target Assignment and the Baseline DDD Estimate
Notes: This figure presents a non-parametric randomization-inference (permutation) test of the composite DDD estimate. Target status is randomly reassigned across individuals and the placebo DDD coefficient recomputed 1,000 times, building the distribution of estimates expected if treatment eligibility carried no signal. The shaded curve is the placebo distribution and the vertical line the observed effect; the placebo distribution centers near zero and the observed estimate lies in its extreme tail, indicating the result is very unlikely to arise by chance. Being distribution-free, the test complements the state-clustered standard errors.

Appendix

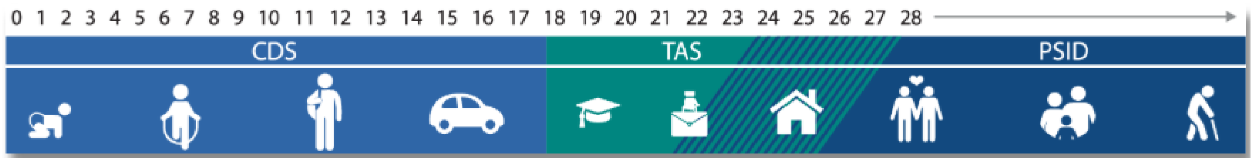

Fig. A1: Time of how CDS, TAS, and PSID map onto children's development.
Notes: This figure illustrates the longitudinal structure of the PSID-CDS-TAS data used in this study. The CDS sampled children born between 1983 and 1997 at ages 0 to 12 at baseline (wave I) and followed them through age 17. Upon turning 18, these same individuals transitioned into the TAS, which tracks them biennially through approximately age 28.

Table A1: State Welfare Reform Implementation Dates

State	Waiver Date	TANF Date	State	Waiver Date	TANF Date
Alabama		11/15/1996	Montana	2/1/1996	2/1/1997
Alaska		7/1/1997	Nebraska	10/1/1995	12/1/1996
Arizona	11/1/1995	10/1/1996	Nevada		12/3/1996
Arkansas	7/1/1994	7/1/1997	New Hampshire		10/1/1996
California	12/1/1992	1/1/1998	New Jersey	10/1/1992	7/1/1997
Colorado		7/1/1997	New Mexico		7/1/1997
Connecticut	1/1/1996	10/1/1996	New York		11/1/1997
Delaware	10/1/1995	3/10/1997	North Carolina	7/1/1996	1/1/1997
District of Columbia		3/1/1997	North Dakota		7/1/1997
Florida		10/1/1996	Ohio	7/1/1996	10/1/1996
Georgia	1/1/1994	1/1/1997	Oklahoma		10/1/1996
Idaho		7/1/1997	Oregon	2/1/1993	10/1/1996
Illinois	11/23/1993	7/1/1997	Pennsylvania		3/3/1997
Indiana	5/1/1995	10/1/1996	Rhode Island		5/1/1997
Iowa	10/1/1993	1/1/1997	South Carolina		10/12/1996
Kansas		10/1/1996	South Dakota	6/1/1994	12/1/1996
Kentucky		10/18/1996	Tennessee	9/1/1996	10/1/1996
Louisiana		1/1/1997	Texas	6/1/1996	11/5/1996
Maine		11/1/1996	Utah	1/1/1993	10/1/1996
Maryland	3/1/1996	12/9/1996	Virginia	7/1/1995	2/1/1997
Massachusetts	11/1/1995	9/30/1996	Washington	1/1/1996	1/10/1997
Michigan	10/1/1992	9/30/1996	West Virginia	2/1/1996	1/11/1997
Minnesota		7/1/1997	Wisconsin	1/1/1996	9/1/1997
Mississippi	10/1/1995	7/1/1997	Wyoming		1/1/1997
Missouri	6/1/1995	12/1/1996			

Notes: This table shows the exact dates adopted either a waiver or the TANF program across the United States. Source: U.S. Department of Health and Human Services.

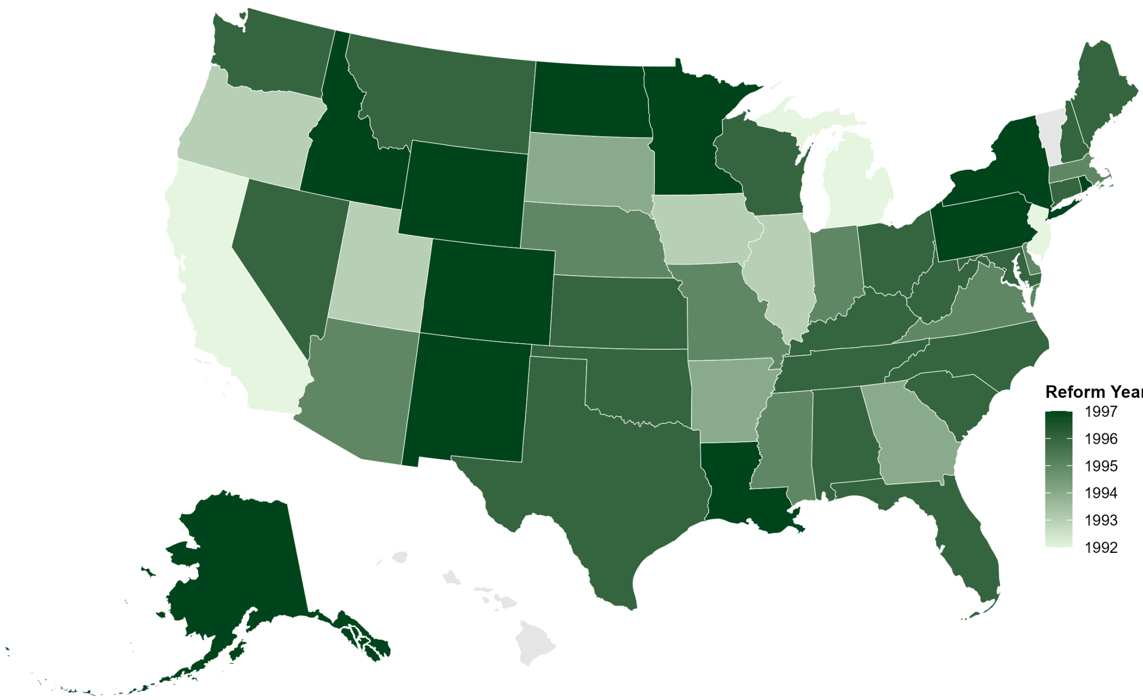

Fig. A2: Welfare Reform Implementation Timing by State
Notes: This figure shows the timing of state-level welfare reform implementation across the United States. Source: U.S. Department of Health and Human Services.

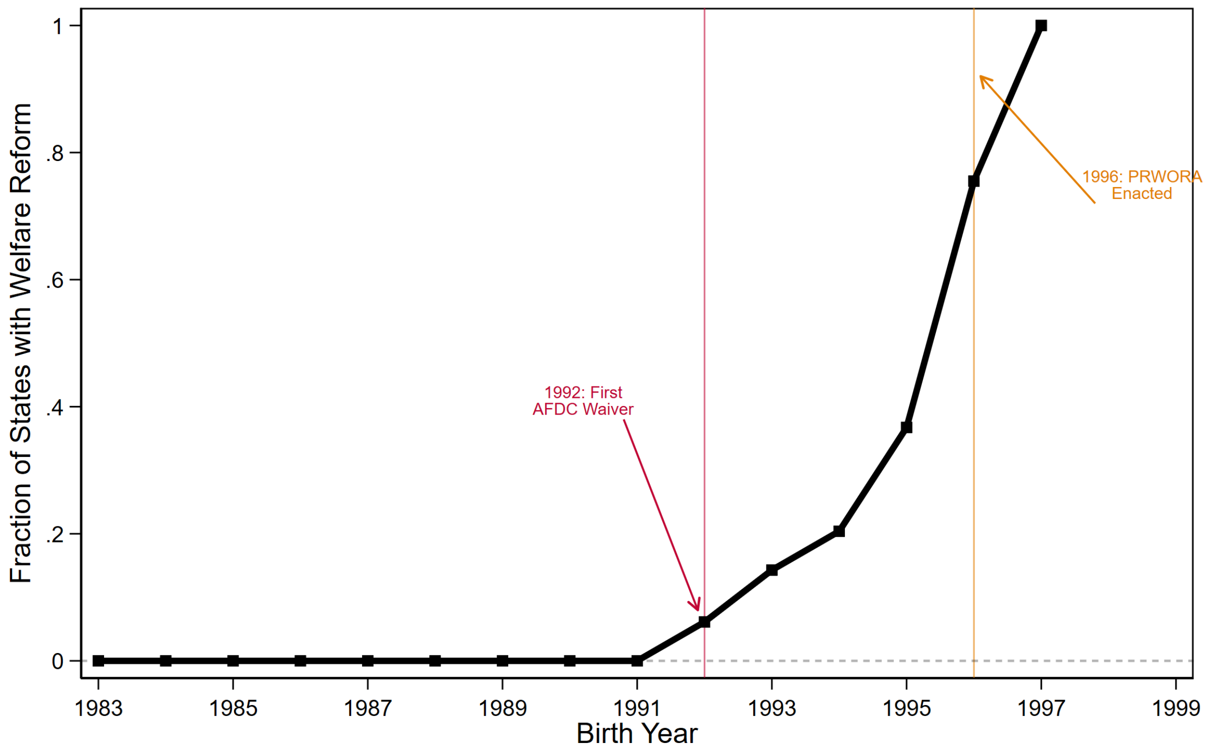

Fig. A3: Staggered Rollout of State Welfare Reform
Notes: This figure shows the staggered rollout of 1990s welfare reform across states, plotting the fraction of states that had implemented reform against child birth year. A state is coded as reformed for a given cohort if the child was born on or after the state's reform adoption date (the earlier of its waiver or TANF implementation). The vertical lines mark the first AFDC waiver (1992) and the enactment of PRWORA (1996).

Table A2: Balance Table

	Control Mean	Target Mean	p-value	SMD
Young Adult Controls				
Age at Survey	23.410	23.572	0.303	0.060
Female	0.477	0.542	0.125	0.130
Nonwhite	0.201	0.626	0.000	0.957
Grandparent Present	0.020	0.050	0.241	0.169
Maternal Controls				
Mother's Age	34.437	32.526	0.067	0.234
< High School	0.098	0.448	0.000	0.854
High School	0.269	0.552	0.000	0.601
Some College	0.543	0.000	0.000	1.542
Pre-Reform State Level Characteristics				
Unemployment Rate	6.260	6.615	0.041	0.264
ln(TANF Benefit)	12.476	12.519	0.749	0.043
ln(SNAP Benefit)	13.088	13.188	0.283	0.123
Poverty Rate	13.310	14.629	0.023	0.364
Minimum Wage	3.759	3.744	0.770	0.028
EITC Rate	0.010	0.019	0.423	0.126

Notes: This table reports weighted summary statistics for the target and control groups. Means are computed using TAS survey weights. The p-values are from two-sample t-tests of equality in means across groups. Standardized mean differences (SMDs) are calculated as the absolute difference in group means divided by the pooled standard deviation, where the pooled standard deviation is defined as the square root of the average of the two group-specific variances. Following common practice, an SMD below 0.10 indicates good covariate balance. Sample weights applied.

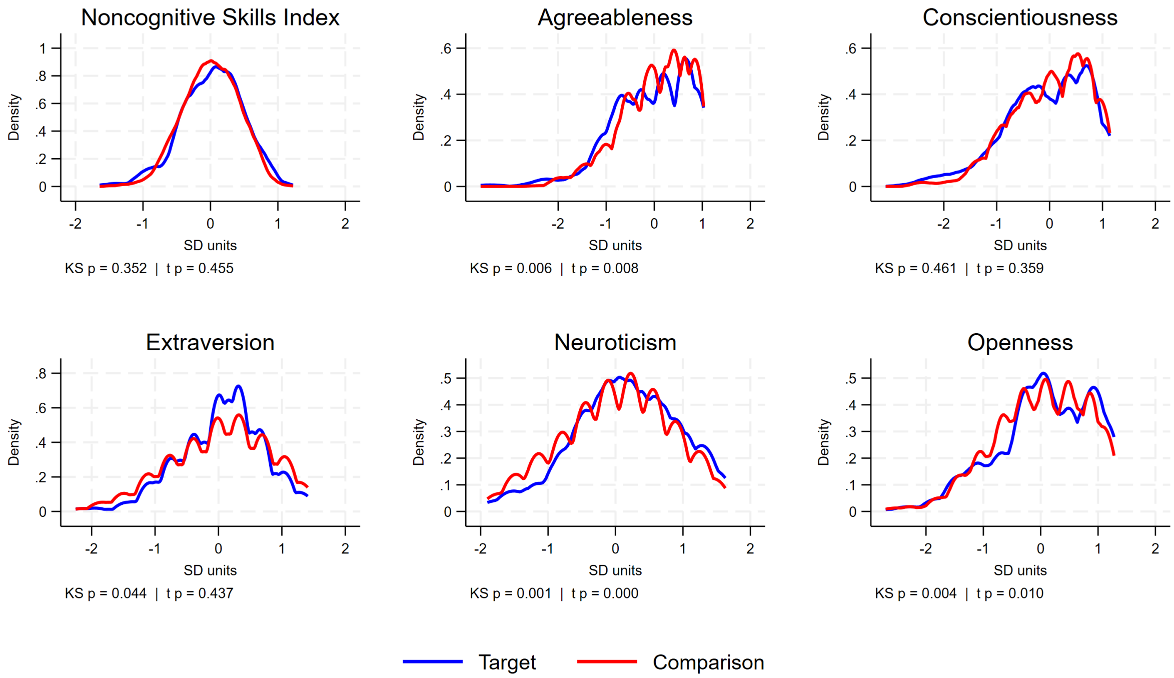

Fig. A4: Distribution of Noncognitive Skills: Target vs. Comparison Children
Notes: This figure plots survey-weighted kernel density estimates of the standardized composite NCS index and each Big Five subscale, comparing target children (born to single mothers with at most a high school education or less) with the comparison group (single mothers with more than high school education or from two-parent household). Distributions are evaluated on a common grid. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of distributions and from a two-sample t-test of equal means between the two groups.

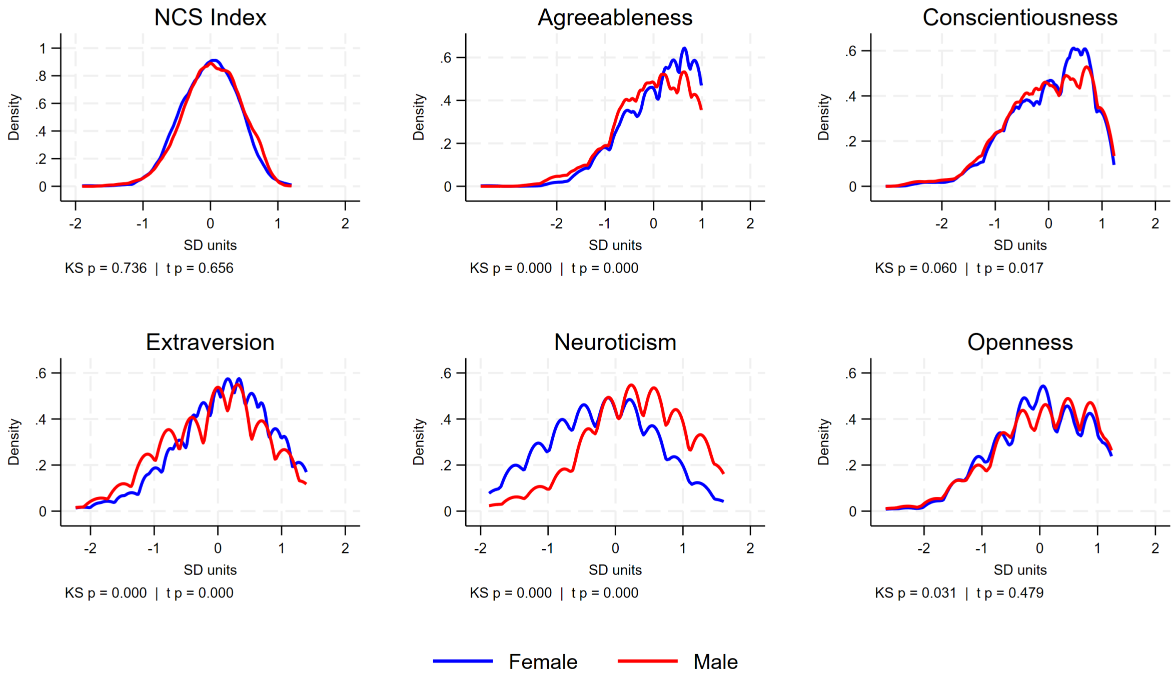

Fig. A5: Distribution of Noncognitive Skills, by Child Gender
Notes: This figure plots survey-weighted kernel density estimates of the standardized composite NCS index and each Big Five subscale, comparing female and male young adults. Distributions are evaluated on a common grid. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of distributions and from a two-sample t-test of equal means between the two groups.

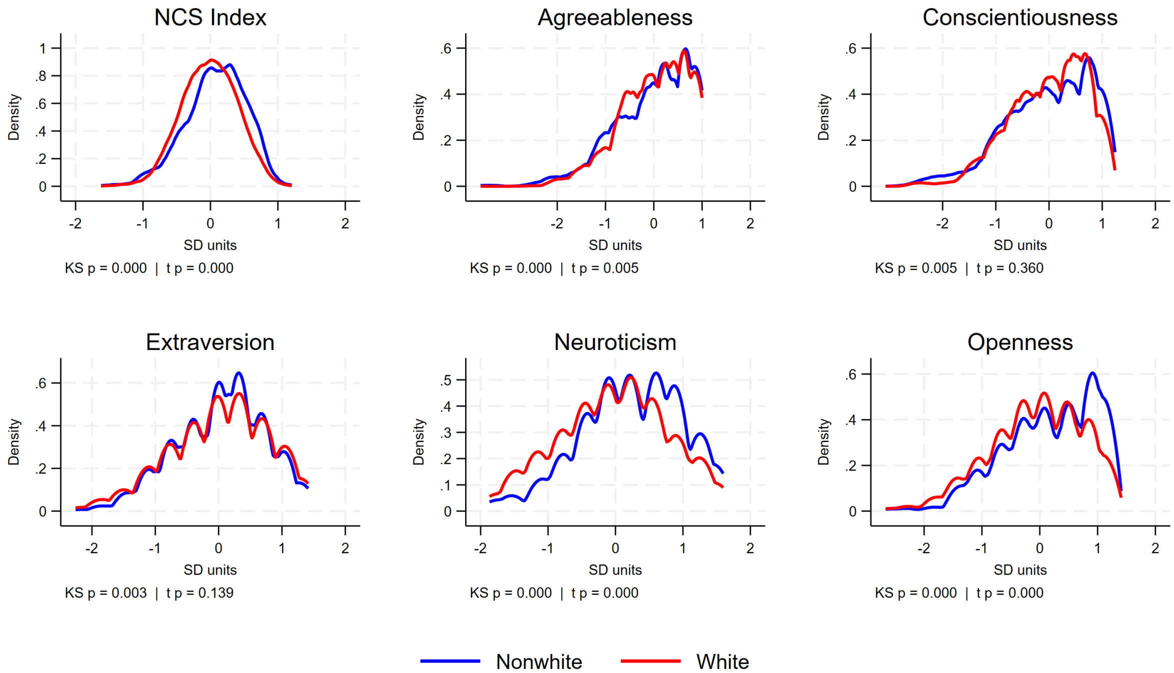

Fig. A6: Distribution of Noncognitive Skills, by Child Race

Notes: This figure plots survey-weighted kernel density estimates of the standardized composite NCS index and each Big Five subscale, comparing Nonwhite and White children. Distributions are evaluated on a common grid. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of distributions and from a two-sample t-test of equal means between the two groups.

Table A3: Reduced Form: Reform Exposure and Young Adult Labor Outcomes

	(1) Earnings (PPML)	(2) Employment (LPM)
Exposure IU-Age 5	0.288*	0.032
	(0.160)	(0.047)
IU-Age 5 × Target	-0.176	0.053
	(0.352)	(0.112)
Observations	2830	2833

Notes: This table reports reduced-form DDD estimates of the effect of early-life welfare-reform exposure on adult labor outcomes. Column (1) models earnings using Poisson pseudo-maximum-likelihood (PPML); column (2) models employment using a linear probability model. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less, and IU-Age 5 × Target is the DDD estimate. Both specifications include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

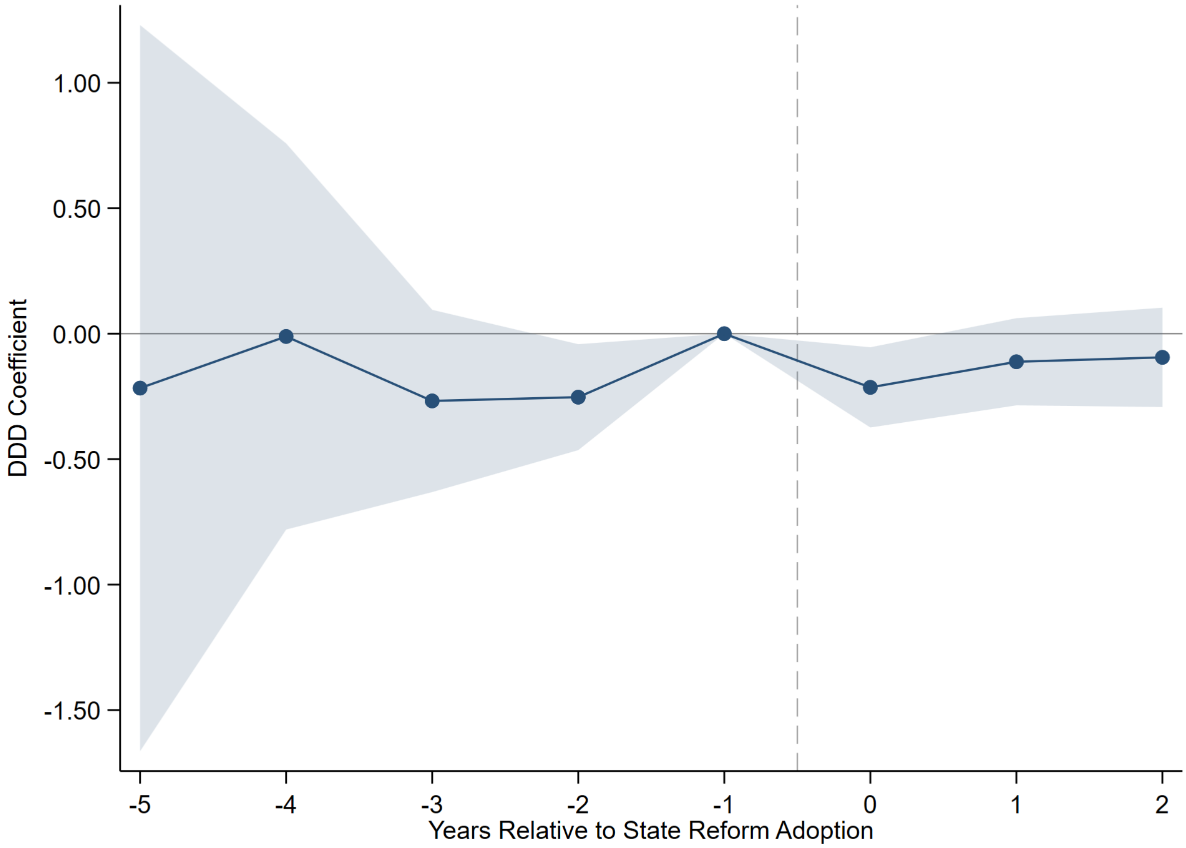

Fig. A7: Event-Study Estimates by Year Relative to State Reform Adoption (Stacked DDD)
Notes: This figure plots the dynamic DDD coefficients from a stacked event-study design for the composite NCS index. Because no state remains untreated within the sample window, each adoption cohort is matched to not-yet-treated states (those adopting later), following Cengiz et al. (2019), and the sub-experiments are stacked with cohort-specific fixed effects. Coefficients are measured relative to the omitted reference year ($t = -1$), with post-adoption years at $t \geq 2$ pooled. The joint test of the pre-adoption coefficients yields $p = 0.275$, supporting parallel trends. Shaded bands are 90% confidence intervals from standard errors clustered at the state level; estimates are survey weighted.

Table A4: Survey Items for the Big Five Personality Scales

Personality Factor	Definition	Survey Items
Agreeableness	The tendency to act in a cooperative, unselfish manner.	Sometimes rude to others *(R)* Forgiving Considerate and kind to others
Conscientiousness	The tendency to be organized, responsible, and hardworking.	Thorough worker Somewhat lazy *(R)* Effective and efficient Completes tasks
Extraversion	An orientation of one's interests and energies toward the outer world of people and things, characterized by positive affect and sociability.	Communicative, talkative Outgoing, sociable Reserved *(R)*
Neuroticism / Emotional Stability	Emotional stability reflects predictability and consistency in emotional reactions, with an absence of rapid mood changes; neuroticism reflects a chronic level of emotional instability and proneness to distress.	Worrier *(R)* Nervous *(R)* Relaxed, able to deal with stress
Openness to Experience	The tendency to be open to new aesthetic, cultural, or intellectual experiences.	Original, comes up with new ideas Values artistic, aesthetic experience Imaginative

Notes: Items are drawn from the PSID Transition into Adulthood self-report personality battery. Definitions follow the American Psychological Association Dictionary of Psychology. Items marked (R) are reverse-coded prior to scale construction so that higher values denote more of the named trait.

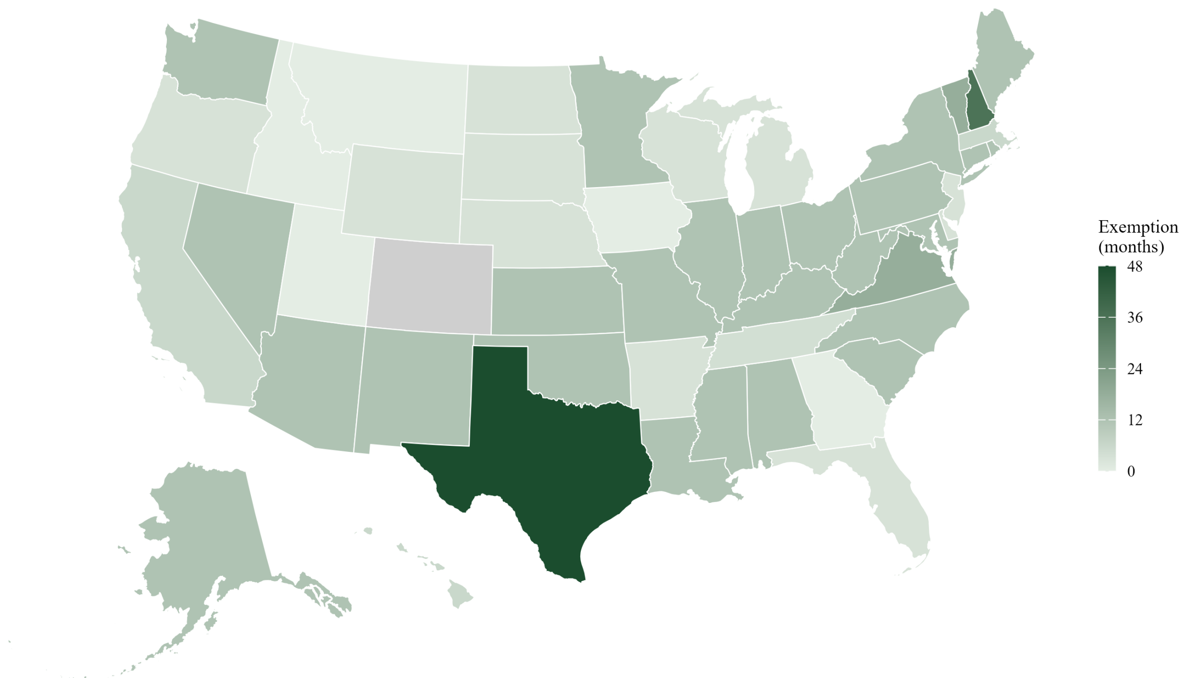

Fig. A8: Age-of-Youngest-Child Work Exemption under TANF by State

Table A5: Age-of-Youngest-Child Work Exemption (AYCE) under TANF, by State

Exemption threshold (age of youngest child at which the parent must participate in work activity), converted to months.

State	TANF Exemption Threshold	Months	State	TANF Exemption Threshold	Months
Alabama	1 year	12	Nebraska	3 months	3
Alaska	1 year	12	Nevada	1 year	12
Arizona	1 year	12	New Hampshire	3 years	36
Arkansas	3 months	3	New Jersey	12 weeks	3
California	6 months	6	New Mexico	1 year	12
Colorado	county	missing	New York	1 year	12
Connecticut	1 year	12	North Carolina	1 year	12
Delaware	13 weeks	3	North Dakota	3 months	3
DC	1 year	12	Ohio	1 year	12
Florida	3 months	3	Oklahoma	1 year	12
Georgia	No	0	Oregon	90 days	3
Hawaii	6 months	6	Pennsylvania	1 year	12
Idaho	No	0	Rhode Island	1 year	12
Illinois	1 year	12	South Carolina	1 year	12
Indiana	1 year	12	South Dakota	12 weeks	3
Iowa	No	0	Tennessee	4 months	4
Kansas	1 year	12	Texas	4 years	48
Kentucky	1 year	12	Utah	No	0
Louisiana	1 year	12	Vermont	18 months	18
Maine	1 year	12	Virginia	18 months	18
Maryland	1 year	12	Washington	1 year	12
Massachusetts	6 months	6	West Virginia	1 year	12
Michigan	3 months	3	Wisconsin	12 weeks	3
Minnesota	1 year	12	Wyoming	3 months	3
Mississippi	1 year	12			
Missouri	1 year	12			
Montana	No	0			

Notes: Age-of-youngest-child work exemption under TANF, from ASPE Table W-2.a (Setting the Baseline: A Report on State Welfare Waivers). The threshold is the age of the youngest child at or below which a parent is exempt from work requirements; higher values denote more protective policies. Thresholds are converted to months (1 year = 12; 13 weeks, 12 weeks, and 90 days = 3). States coded No exemption are set to 0 (least generous). Colorado sets exemptions at the county level and is coded missing. This threshold is the exempt-months moderator used in the AYCE heterogeneity analysis.

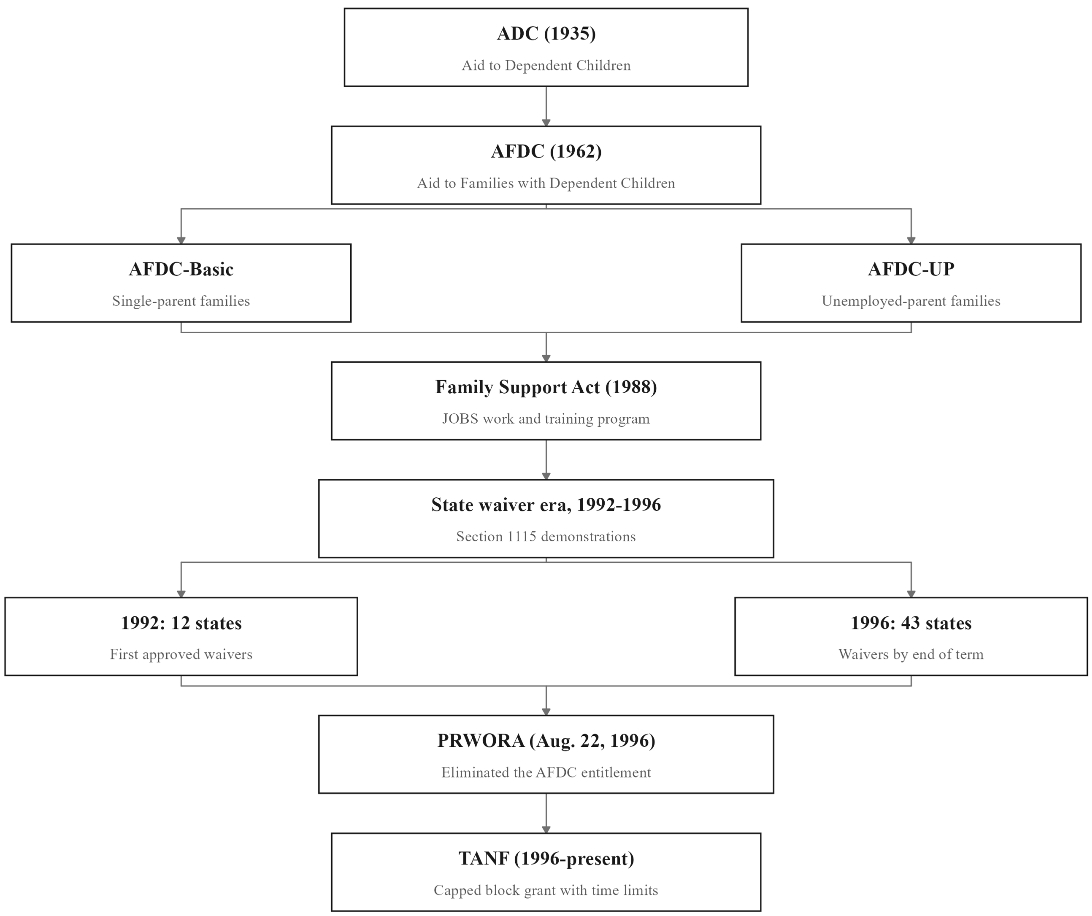

Fig. A9: Welfare Reform Genealogy Plot

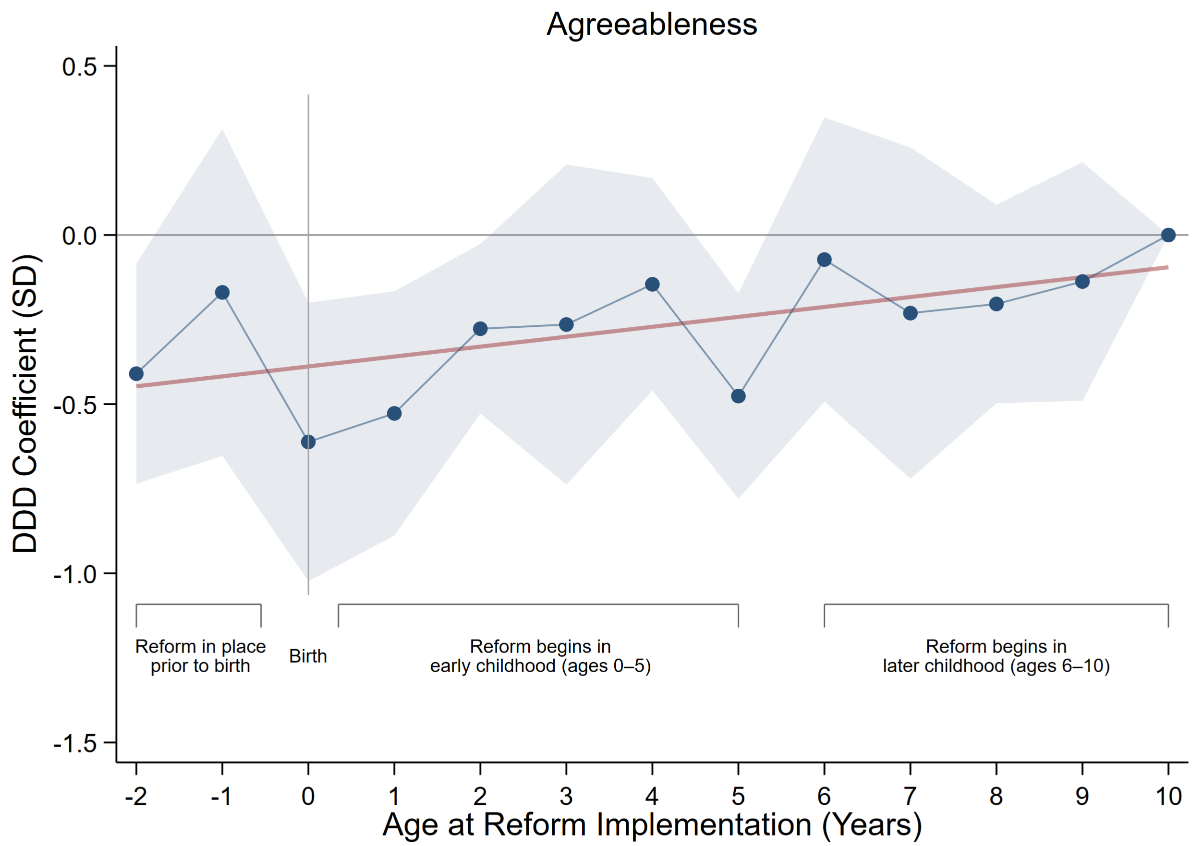

Fig. A10: Effect of Welfare Reform by Age at Exposure: Agreeableness

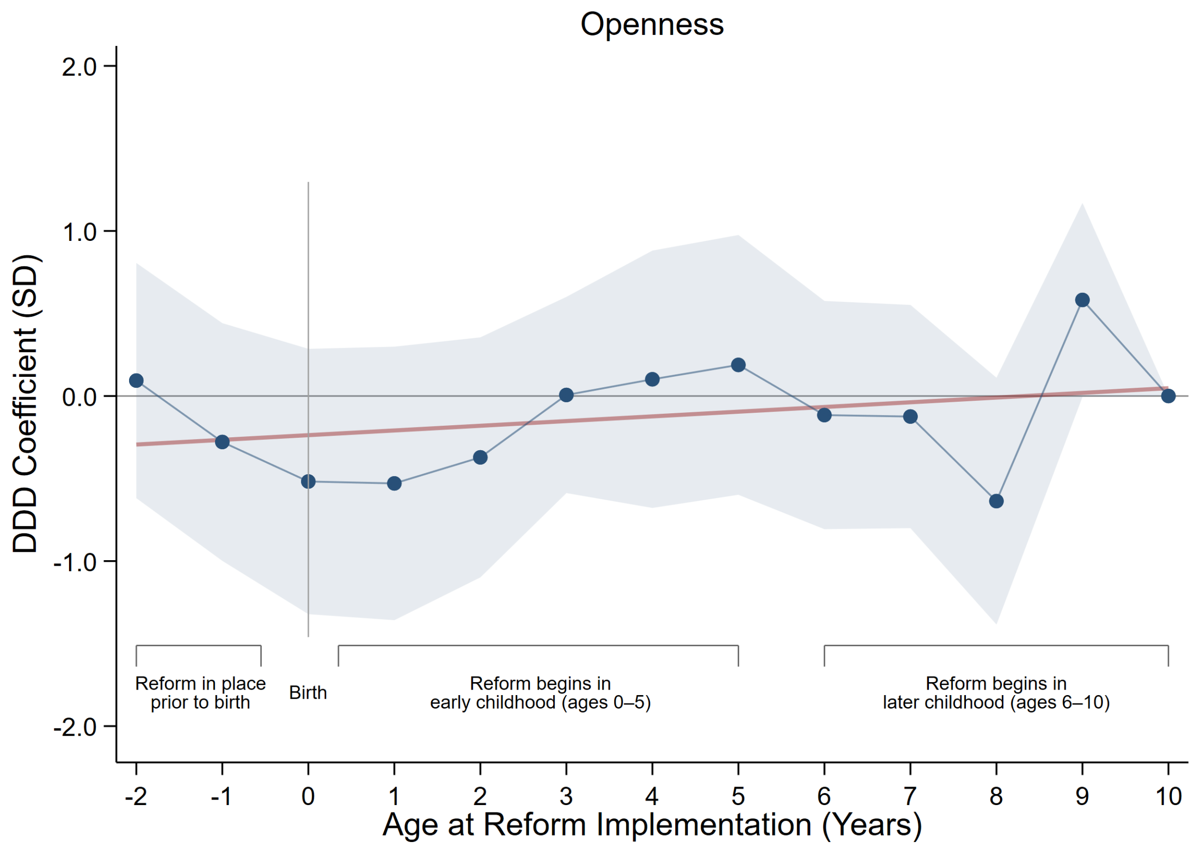

Fig. A11: Effect of Welfare Reform by Age at Exposure: Openness

Technical Appendix

Appendix A. Construction of the NCS Index

Let i index individuals and j index the noncognitive items derived from the Big Five battery. All items are reported on a 1 to 4 scale. For some items a higher raw score corresponds to a worse trait (for example, "rude," "lazy," "nervous"); for others a higher score corresponds to a better trait (for example, "kind," "forgiving").

A.1. Reverse-coding of negatively framed items

Let r_{ij} denote the raw response of individual i to item j. Let N be the set of negatively framed items (rude, lazy, reserved, worries, nervous) and P the remaining positively framed items. I first map all items to a common direction in which higher values indicate more favorable NCS,

$$x_{ij} = \begin{cases} 5 - r_{ij} & \text{if } j \in N, \\ r_{ij} & \text{if } j \in P. \end{cases} \tag{A1}$$

By construction x_{ij} remains on the 1 to 4 scale, with larger values uniformly interpretable as more favorable traits.

A.2. Control-group standardization (Kling-style z-scores)

Let $D_i = 1$ if child i is in the target group (a mother with at most a high school education) and $D_i = 0$ if in the comparison group. Following Kling et al. (2007), I standardize each item using the comparison group only, so that effects are expressed in standard-deviation units relative to the untreated distribution. For each item j, I compute the comparison-group mean and standard deviation,

$$\mu_j = E[x_{ij} \mid D_i = 0], \qquad \sigma_j = SD[x_{ij} \mid D_i = 0], \tag{A2}$$

and define the standardized score for individual i on item j as

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{A3}$$

so that $z_{ij} = 1$ means child i lies one comparison-group standard deviation above the comparison-group mean on trait j.

A.3. Overall noncognitive index

The overall NCS index is the simple average of the 15 standardized items. Let J_i denote the set of non-missing items for individual i. The index is

$$NCS_i = \frac{1}{|J_i|} \sum_{j \in J_i} z_{ij} \tag{A4}$$

If all items are observed then $|J_i| = 15$; otherwise the mean is taken over the available items. By construction, among comparison-group children the index has mean approximately zero and variance close to one, so every treatment coefficient reads as a standard-deviation change in overall NCS relative to the comparison group.

Appendix B. Construction of the Welfare-Reform Exposure Variable

The treatment variable measures the share of a child's earliest developmental window spent under post-reform welfare rules. This appendix documents its construction for the primary in-utero to age-five window; the age 6 to 12 window is built analogously with an 84-month denominator.

B.1. Timing anchors

For each child I observe the birth year and birth month. I date conception to nine months before birth, so the conception month is the birth month minus nine (rolling into the prior calendar year when the difference is non-positive), with the conception year decremented accordingly. The exposure window runs from conception through the fifth birthday, a span of 69 months (9 months in utero plus 60 months of life), and all exposure shares use 69 months as the denominator.

B.2. Assigning reform dates to states

Each child is assigned to a state, the state of residence in the 1997 CDS. To that state I attach two policy dates recorded to the month, the implementation date of its AFDC Section 1115 waiver where one was granted, and the implementation date of its TANF program under PRWORA. Reform timing is the earlier of the two, so a state is treated from the first date it departed from the pre-reform AFDC regime.

B.3. Months of exposure under each regime

For the waiver regime I count the window-months falling on or after the waiver date. Let the waiver be implemented in month m_w of year y_w, and let conception occur in month m_c of year y_c. The waiver-exposed months are

$$E_i^w = 12(y_w - y_c) + (m_w - m_c), \tag{B1}$$

counting how much of the 69-month window lies at or after the waiver date. This quantity is bottom-coded at 0 (children whose window closed before the waiver) and top-coded at 69 (children conceived entirely under the waiver). Because a state's waiver was superseded once TANF took effect, waiver exposure is additionally capped at the waiver's own duration, the number of months between the waiver and TANF dates, so months after TANF are not double-counted. TANF-exposed months E_i^T are computed by the identical formula using the TANF date, bottom-coded at 0 and top-coded at 69.

B.4. Interview-date correction

Some children had not reached age five at the 1997 CDS interview, so their window was not yet complete. For these children I cap exposure at the share of the window actually elapsed by the interview date, subtracting any excess so that exposure never exceeds the portion of the window a child had lived. Children already past age five completed the full window and are unaffected.

B.5. Total exposure and the treatment interaction

Each regime's exposure is converted to a share by dividing by 69, and total early-life exposure is the sum of the waiver and TANF shares, capped at one,

$$RE_i^{\text{IU-5}} = \min\left\{\frac{E_i^w + E_i^T}{69},\ 1\right\}. \tag{B2}$$

A value of 0 denotes a child whose entire window predated any reform in their state, and a value of 1 a child who spent the whole window under post-reform rules. The continuous-dose DDD treatment is the interaction of this exposure share with target status,

$$RE_i^{\text{IU-5}} \times T_i \tag{B3}$$

which is the coefficient of interest throughout the paper.

Appendix C. Robustness Methods

This appendix sets out the three methods used to probe the robustness of the baseline triple-difference estimate, the stacked estimator that guards against negative weighting under staggered adoption, the permutation test that provides distribution-free inference, and the Oster (2019) procedure that bounds the scope for omitted-variable bias.

C.1. Stacked Triple-Difference Estimator

Under staggered adoption with heterogeneous effects, the two-way fixed-effects estimator can form forbidden comparisons in which already-treated units serve as controls for later-treated ones, delivering a weighted average of treatment effects in which some weights are negative (Goodman-Bacon, 2021; de Chaisemartin and d'Haultfoeuille, 2020). Because every state in my sample adopts reform between 1992 and 1996, no never-treated group is available. The stacked design (Cengiz et al., 2019) circumvents this by building a set of clean sub-experiments, each with its own not-yet-treated controls, and estimating the baseline specification on the pooled stack.

Step 1. Define adoption cohorts. Let c index the calendar years in which states adopt reform, $c \in \{1992, \ldots, 1996\}$. Each adoption year defines one sub-experiment, indexed by e.

Step 2. Assign treated and control units. For sub-experiment e tied to adoption year c, the treated group comprises children born in states adopting in year c. The control group comprises children in not-yet-treated states, those adopting strictly after c. States adopting before c are excluded, since they are already treated and would reintroduce forbidden comparisons.

Step 3. Trim to a common event window. Within each sub-experiment I retain birth cohorts in a symmetric window around the adoption year, from five cohorts before to four after, so every sub-experiment contributes a comparable range of relative cohorts and thin boundary cells are excluded.

Step 4. Stack the sub-experiments. The trimmed sub-experiments are appended into a single dataset. A child near a cohort boundary may appear in more than one sub-experiment under a different reference each time; this is a feature of the design, not double counting, because identification proceeds within each e.

Step 5. Estimate with sub-experiment-specific fixed effects. On the stacked sample I estimate the same triple difference as in the main text, with every fixed effect interacted with the sub-experiment index e,

$$NCS_{isce} = \delta_1 RE^{\text{IU-5}}_{isc} + \delta_2 T_i + \delta_3 \left(RE^{\text{IU-5}}_{isc} \times T_i\right) + \beta X_{isce} + \alpha_{se} + \gamma_{ce} + \eta_{te} + \theta_{Te} + \varepsilon_{isce} \quad \text{(C1)}$$

where α_{se}, γ_{ce}, η_{te}, and θ_{Te} are state, birth-cohort, survey-year, and target fixed effects specific to sub-experiment e. The estimator is identical to the baseline; only the sample and the indexing of the fixed effects change. Because these fixed effects confine every contrast to within a single sub-experiment, treated cohorts are compared only against not-yet-treated states, and the forbidden comparisons that generate negative weights cannot arise. The coefficient δ_3 recovers the triple-difference effect, and standard errors are clustered on state.

C.2. Permutation Inference

Treatment varies at the state level and the number of clusters is moderate, so the asymptotic approximation underlying clustered standard errors may be unreliable. I complement conventional inference with a non-parametric randomization test (Ferrara et al., 2012) that asks how often an effect as large as the observed one would arise if treatment eligibility carried no signal.

Step 1. Record the observed estimate. Estimate the baseline specification and store the coefficient on the interaction, $\hat{\beta}$.

Step 2. Construct a placebo treatment. For replication r, randomly reassign target status across individuals, drawing a placebo indicator $T_i^{(r)}$ that preserves the share of treated units but severs any true link to family background. Form the placebo interaction $RE_{isc}^{\text{IU-5}} \times T_i^{(r)}$.

Step 3. Re-estimate and store the placebo coefficient. Re-estimate the baseline specification with the placebo interaction in place of the true one, and store the coefficient $\hat{\beta}^{(r)}$.

Step 4. Repeat to build the null distribution. Repeat Steps 2 and 3 for $R = 1{,}000$ replications, yielding an empirical distribution $\{\hat{\beta}^{(r)}\}$ that approximates the distribution of the estimate under the null of no effect. Because target status is assigned at random, this distribution centers on zero.

Step 5. Compute the randomization p-value. The two-sided permutation p-value is the share of placebo estimates at least as large in magnitude as the observed coefficient,

$$p = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}\left(|\hat{\beta}^{(r)}| \geq |\hat{\beta}|\right) \tag{C2}$$

Because it is built entirely from the data through random relabeling, the test is distribution-free and makes no appeal to large-sample theory. An observed estimate lying in the extreme tail of the placebo distribution, with a correspondingly small p, indicates the effect is very unlikely to have arisen by chance.

C.3. Bounding Omitted-Variable Bias (Oster, 2019)

Selection on unobservables cannot be ruled out by controls alone. Oster (2019) formalizes the intuition that if adding observed controls moves the coefficient little while raising the R-squared, then unobservables of comparable explanatory power would also move it little. The procedure uses the movement of the coefficient and the R-squared between an uncontrolled and a controlled regression to bound the bias.

Step 1. Estimate the uncontrolled and controlled regressions. Let $\tilde{\beta}$ and $\tilde{R}$ be the coefficient and R-squared from a regression with only the fixed effects, and $\dot{\beta}$ and $\dot{R}$ the coefficient and R-squared from the controlled regression that adds the full set of covariates.

Step 2. Set the maximum R-squared. Specify $R_{\max}$, the R-squared a regression on both observed and unobserved determinants would attain. Following common practice I set $R_{\max} = \min\{1.3\,\dot{R},\ 1\}$, allowing unobservables to explain up to thirty percent more variation than the observed controls.

Step 3. Compute the bias-adjusted coefficient. Under equal selection on observables and unobservables ($\delta = 1$), the bias-adjusted coefficient is approximately

$$\beta^* \approx \dot{\beta} - (\tilde{\beta} - \dot{\beta})\frac{R_{\max} - \dot{R}}{\dot{R} - \tilde{R}} \tag{C3}$$

The identified set for the true effect is the interval bounded by the controlled coefficient $\dot{\beta}$ and the bias-adjusted β^*. An identified set that excludes zero indicates the effect survives proportional selection on unobservables.

Step 4. Compute the critical δ. Alternatively, fix the target at $\beta = 0$ and solve for the δ required to attribute the entire effect to unobservables. This δ measures how large selection on unobservables would have to be, relative to selection on the observed controls, to drive the coefficient to zero. A value greater than one means unobservables would need to be more important than the full set of observed controls, an implausibly strong requirement; a negative suggests that omitted-variable bias, if proportional to selection on observables, would strengthen rather than weaken the estimated effect.

Step 5. Interpret the bounds. I report both objects for each outcome, the identified set from equation (C3) and the critical δ for $\beta = 0$. Coefficient stability across the addition of controls, an identified set excluding zero, and a large or negative critical δ together indicate that the estimate is unlikely to be an artifact of unobserved confounding.